Bridging the Gap of Missing Modalities: Common-Specific Knowledge Learning Network for Multimodal Remote Sensing Image Classification

Simin Zhu, Zhaohui Xue, *Senior Member, IEEE*, and Hao Wu

***Abstract*—Multimodal learning has attracted significant attention in the field of remote sensing image classification due to its superior performance. Although multimodal data are often available during training, certain modalities may be unavailable during inference in real-world applications, posing a risk of performance degradation. To this end, we propose a novel common-specific knowledge learning network (CSKLNet) to bridge the gap of missing modalities. CSKLNet consists of two learning stages. First, complete multimodal knowledge is distilled from the teacher into the student under missing modalities, with the Wasserstein distance adopted to alleviate representation heterogeneity by minimizing feature distribution discrepancies. Second, two parameter-efficient fine-tuning (PEFT) adapters are inserted into the distilled student model to further exploit specific information from the available modalities. Compared with existing methods, CSKLNet not only inherits the common information from complete modalities but also fully leverages the available modality-specific information, thereby maintaining robust performance with missing modalities. Extensive experiments conducted on four distinct datasets indicate the advanced performance of the proposed method compared with state-of-the-art (SOTA) methods, with overall accuracy (OA) improvements of 1.34%–13.34%, 0.35%–3.54%, 0.55%–17.65%, and 0.17%–11.91% for hyperspectral image (HSI)–light detection and ranging (LiDAR) Houston2013, Trento, HSI–multispectral (MS) Houston2013, and QAI-Mangrove, respectively. The source code will be available at: https://github.com/ZhaohuiXue/CSKLNet**

***Index Terms*—Classification, fine-tuning, knowledge distillation, missing modalities, multimodal, remote sensing.**

I. Introduction

REMOTE sensing image classification plays an increasingly prominent role in a wide range of Earth observation (EO) tasks, such as urban development planning [1], forestry monitoring [2], and disaster response [3]. With the rapid development in EO technology, diverse types of EO data, such as hyperspectral image (HSI), multispectral (MS), and light detection and ranging (LiDAR), have become publicly available to the research community. Classification models that leverage multiple modalities have demonstrated substantial advantages and sparked considerable attention in this field [4], [5], [6], [7], [8], [9]. Although these methods have achieved compelling success, they generally rely on the assumption that all modalities are available, which is often difficult to guarantee in real-world applications due to factors such as sensor failures [10] or weather conditions [11], thereby rendering certain modalities unavailable at the test phase. When conventional multimodal classification models encounter missing modalities, their performance typically degrades significantly. Therefore, enabling models to remain robust under missing modalities is of great significance.

To date, several works have focused on addressing multimodal learning under missing modalities, and most methods can be classified into three types: generation-based, representation-based, and distillation-based methods. Naturally, when considering missing modalities, generating the absent modalities emerges as a straightforward strategy, giving rise to a large body of generation-based methods [12], [13], [14], [15], [16]. These methods leverage the available modalities to synthesize the missing ones via generative models, thereby forming complete multimodal input compatible with conventional multimodal models. Although these approaches can partially alleviate the performance degradation induced by missing modalities, generative models substantially increase model complexity and computational demands [17], and training generators capable of producing high-quality missing modalities remains a challenging problem [18].

Therefore, many studies introduce specific constraints to enable models to learn semantic consistency between different modal representations, i.e., modality-invariant feature. These approaches can be referred to as representation-based methods. These methods [19], [20], [21], [22], [23] assist the model in learning how to leverage feature from available modalities to approximate feature of missing modalities or incorporate random modality dropout strategy during training to facilitate learning of intermodal feature similarity and corre-

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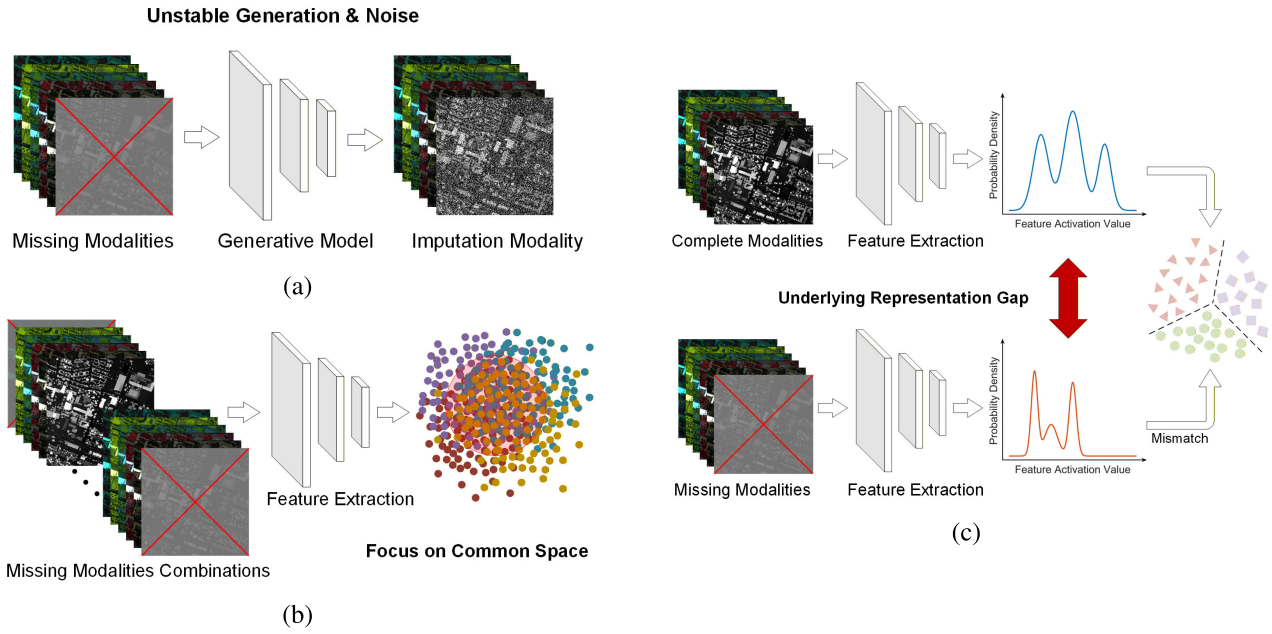


Fig. 1. Illustration of the limitations in existing works. (a) Limitation of modality generation. Instability and extra noise may be introduced by data generation. (b) Limitation of common space. Focusing on extracting modality-invariant feature in a common space may overlook modality-specific information. (c) Representation discrepancy. High-level knowledge learning remains insufficient to bridge representation discrepancy under missing modalities.

lation. Although these methods can avoid the instability and complexity of generation-based methods, they primarily focus on capturing modality-invariant feature, while overlooking the specific information possessed by each modality, which is a crucial factor in multimodal remote sensing classification [9].

Unlike generation- or representation-based methods, many studies modify the training architecture to accommodate missing modalities, where distillation-based methods have emerged as the predominant paradigm. Knowledge distillation [24], as a transfer learning technique, has been widely applied to this problem. Specifically, the student mitigates performance degradation under missing modalities by learning multimodal knowledge from the teacher. Rich and complete multimodal knowledge, such as relations and semantics, can be transferred by matching intermediate-level features or output-level logits [25], [26], [27], [28]. To further improve student performance, several approaches [29], [30], [31], [32], [33] employ hybrid distillation that combines both features and logits. Such a strategy effectively integrates multiple forms of knowledge to achieve more precise knowledge transfer. While these methods have achieved progressive results, they primarily focus on extracting diverse forms of knowledge from network representations for transfer, lacking effective addressing of the discrepancies among representations that underlie knowledge, and also pay insufficient attention to modality-specific information.

The above-mentioned methods have achieved excellent performance, there are still several limitations.

1) Generating missing modalities to complete multimodal inputs is inherently unstable and complex. For highly heterogeneous remote sensing data, the quality of generated modalities is difficult to guarantee and may introduce extra noise that impairs the primary task, as shown in Fig. 1(a). Consequently, avoiding reliance on data generation for handling missing modalities deserves consideration.
2) Most methods aim to establish a common latent space, thereby learning intermodal common information to address missing modalities, while giving limited attention to modality-specific information, as shown in Fig. 1(b), which is critical for multimodal remote sensing classification [9].
3) Existing methods mainly constrain high-level information, such as semantic consistency and attention maps, which are derived from the underlying representations [34]. When a substantial representation gap exists between complete and missing modalities, high-level constraints alone are insufficient to bridge the gap, resulting in discriminative feature misalignment and persistent decision boundary mismatch, as shown in Fig. 1(c).

To address the aforementioned issues, we propose a common-specific knowledge learning network (CSKLNet) for multimodal remote sensing image classification with missing modalities. CSKLNet employs a two-stage learning strategy. First, common intermodal knowledge is transferred from a pretrained multimodal model to the student model via distribution alignment distillation (DAD). Second, the student model is further fine-tuned using adapters, enabling the model to better exploit modality-specific information from the available modalities. Specifically, we effectively address the feature discrepancies between complete and missing modalities through distribution alignment with the Wasserstein distance and further incorporate output logits to impose reliable constraints on the student, thereby transferring the dark knowledge embedded in predictions. Compared with other methods for handling

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missing modalities that only consider common information between modalities, CSKLNet integrates common knowledge of complete modalities and specific knowledge of preserved modalities through two-stage learning, further improving the model's classification performance under missing modalities.

The main contributions of this work are summarized as follows.

1) To effectively learn the common and specific information, we propose a novel CSKLNet with a two-stage learning strategy for multimodal remote sensing image classification under missing modalities.
2) A Wasserstein distance-based distribution alignment strategy is developed, leveraging its ability to capture the geometric structure of feature distributions and provide reliable discrepancy estimation under distribution shifts, making it particularly suitable for aligning highly heterogeneous representations, thereby facilitating common knowledge transfer.
3) Fine-tuning adapters are employed to compensate for the underutilization of modality-specific information during distillation. Through adaptive refinement of the distilled representations, modality-specific characteristics are incorporated into the distilled common knowledge, yielding more comprehensive feature representations.

II. Related Works

A. Knowledge Distillation

Knowledge distillation aims to transfer representational capabilities from a large model to a resource-constrained small model, also known as a teacher–student architecture [35]. In general, based on the knowledge transferred, distillation can be broadly categorized into three types: response-based distillation, feature-based distillation, and relation-based distillation. Response-based distillation typically refers to transferring the neural responses from the teacher model's final output layer, usually achieved by matching the soft output logits [24]. While response-based distillation is straightforward, it relies on the output of the last layer and cannot handle supervisory signals from intermediate layers of the teacher. With the development of representation learning, researchers began to focus on intermediate feature-level distillation. FitNets [36] was the first to introduce intermediate representations of the teacher to supervise student learning, the studies [37] using various types of attention at different representation levels to transfer knowledge and [38] bridging teacher–student networks through feature embedding. Furthermore, relation-based distillation further introduces relationships between different layers or data samples, and transfers it by minimizing similar activations [39], similarity probability distributions [40], and intersample relationships [41]. In summary, these works aim to extract various types of knowledge from different locations and explore how to effectively transfer them, which is the key to knowledge distillation.

B. Adapter-Based Fine-Tuning

An adapter is a representative parameter-efficient fine-tuning (PEFT) method, typically inserting an adapter into a pretrained model. During fine-tuning, most parameters of the pretrained model remain frozen, with only additional module parameters updated, effectively reducing the demand for storage and computational resources. In general, an adapter consists of a down projection layer, a nonlinear activation function, and an up projection layer [42]. Due to excellent generalization ability, adapter fine-tuning (AFT) has achieved great success in computer vision [43] and multimodal learning [44]. In remote sensing, Hu et al. [45] proposed Airs, introducing two independent adapter structures that surpass full fine-tuning on multiple remote sensing vision tasks. Ma et al. [46] introduced a multimodal adapter that fine-tuning the image encoder of SAM, effectively utilizing the model's general knowledge of multimodal data. In general, these different AFT techniques can all bring certain improvements to their respective tasks.

C. Multimodal Learning With Missing Modalities

Traditional multimodal models suffer severe performance degradation when encountering missing modalities. To address this challenge, researchers have proposed various methods, which can be broadly categorized into generation-based methods, representation-based methods, and distillation-based methods.

Generation-based methods utilize existing modalities to synthesize missing modalities through generative models, allowing traditional multimodal models to be effectively applied. Tran et al. [13] used a cascaded residual autoencoder (AE) to progressively generate missing modalities. Recent work [47] has shown that generative adversarial networks (GANs) are superior to AE in modality generation. Bischke et al. [14] used GANs to generate deep data, effectively improving the segmentation performance. Kang et al. [48] constructed cross-modal GANs to generate long-term time-series data, thereby imputing missing values. Liu et al. [49] proposed a multimodal-temporal fusion framework based on intermodality mapping for missing image simulation. Although the above methods can alleviate the performance degradation caused by missing modalities, due to the complexity of data generation [17], these methods are often unstable and may introduce additional noise that affects downstream tasks.

Representation-based methods enable models to learn semantically consistent representations by imposing various operations or specialized constraints at the representation level. For example, iCmSC [21] employs deep canonical correlation analysis (CCA) to maximize cross-modal correlations among incomplete multimodal data, thereby learning a more consistent shared subspace representation and improving clustering performance. mmFormer [19] bridges Transformer and CNN to establish long-distance dependencies between different modalities, enabling the extraction of modality-invariant representation. ShaSpec [20] uses adversarial learning to model and fuse shared and specific features between modalities to deal with missing modalities. In remote sensing, DiffAlignOp [50]

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introduces the metasensor theory, which utilizes prototypes to preserve sensor-invariant feature characteristics; MetaRS [51] employs a metamodal-aware module to extract modality-invariant features for missing modality feature recovery. While these methods have achieved promising results in their respective fields, most of them primarily focus on intermodal consistency information.

Distillation-based methods transfer complete multimodal information from the teacher to the student with missing modalities input through knowledge distillation. For example, Wang et al. [29] proposed ProtoKD, which improves segmentation performance under missing modalities by capturing intraclass and interclass relationships. CLKD [25] combines contrastive learning and knowledge distillation to capture and transfer semantic consistency across modalities. PMKL [30] leverages regularization and contrastive learning to transfer the teacher's rich structural information, thereby enhancing the segmentation performance of a single modality. In remote sensing, MSHNet [31] uses joint adaptation distillation to preserve intermodal interaction in the teacher, enabling the student to reconstruct complete modality-shared features. DGDNet [26] transfers knowledge by matching feature diversity to overcome missing modalities and combines a regularization strategy to alleviate the imbalanced multimodal training. While the above methods have made convincing progress, they have two limitations: one is that these methods emphasize transferring relationships between samples or modalities while neglecting the substantial discrepancies in feature distributions between complete and missing modalities. Another is that most approaches consider only the common information among complete modalities, overlooking the valuable characteristics of the preserved modalities.

III. Methods

A. Overview

In this section, we introduce our proposed method for addressing missing modalities in remote sensing image classification tasks, the overall framework of which is shown in Fig. 2. This method consists of two stages: distillation for common knowledge and fine-tuning for specific knowledge. Specifically, to address the significant heterogeneity between teacher and student representations caused by missing modalities, we propose modeling the intermediate representations using a Gaussian distribution, minimizing the Wasserstein distance between the distributions, and further transferring the rich dark knowledge from the teacher through logits distillation. To more fully utilize the specific information inherent in the preserved modalities, we perform AFT on the student model after distillation. Through the two-stage learning above, the student model learns complete multimodal common knowledge from the teacher, while fully utilizing the specific information inherent in the available modalities, thus improving the model's classification performance under missing modalities.

In summary, we will first introduce the teacher and student model architecture of our method. Next, we will introduce the distillation method for transferring complete multimodal common knowledge, as well as AFT to facilitate learning of modality-specific information. Finally, we will discuss the training details of the method.

B. Model Architecture

Multimodal remote sensing classification models have attracted considerable attention in this field due to their powerful performance. MDL_RS [5] has been proven to have favorable classification capability and generality. Therefore, we adopt it as the backbone architecture of the multimodal network, i.e., the teacher network. For different modalities, independent encoders are used to extract modal features, which are then fused to form a common feature and subsequently passed to a classifier for classification. Note that in this study, we chose middle fusion as the multimodal fusion way. As for the student model, this study considers a single-modality input; thus, only one encoder is required to extract features for subsequent processing. In addition, we propose two additional designs to better address multimodal classification with missing modalities, hyperspectral encoder, and feature weighting (FW), respectively. These two designs will be described in detail below.

1) Hyperspectral Encoder: HSI typically contains hundreds of contiguous spectral bands, providing rich information about land cover [52], [53]. Effectively extracting discriminative information from the numerous spectral bands is crucial [54]. However, conventional 2-D convolution mainly operates on spatial dimensions and has limited capability to model spectral information. Meanwhile, HSI also contains spatial information, making it important to jointly exploit spectral and spatial features [55]. To this end, we employ a collaborative encoder that sequentially integrates 3-D and 2-D convolutions. Specifically, 3-D convolution captures spectral correlations, while the subsequent 2-D convolution further extracts high-level spatial representations, enabling complementary exploitation of spectral and spatial information.

First, to meet the input requirements of 3-D convolution, an additional channel dimension (from $S \times H \times W$ to $1 \times S \times H \times W$) is added to the original HSI. Then, batch normalization (BN) is applied along the newly added dimension to stabilize the training

$$X_{\mathrm{BN}}^{1\times S\times H\times W} = \mathrm{BN}\left(X^{1\times S\times H\times W}\right). \tag{1}$$

Next, a dilated convolution with a kernel size of $3 \times 1 \times 1$ and a dilation rate of 3 is applied to the spectral dimension (S). This allows for the perception of more distant band dependency while maintaining a low parameter count, capturing macroscopic spectral relationships across bands. Simultaneously, the newly added channel dimension is increased from 1 to 64, enabling the compressed information to unfold in a higher dimensional channel space, thus avoiding the loss of important feature. To further integrate the spectral context, a large-kernel convolution (kernel size $S/3 \times 1 \times 1$) covering the entire spectral axis is applied to the feature obtained in the previous step, completely aggregating the spectral dimensions into a single spectral value and performing squeeze. The number of channel is also expanded to 256 to enhance the spectral

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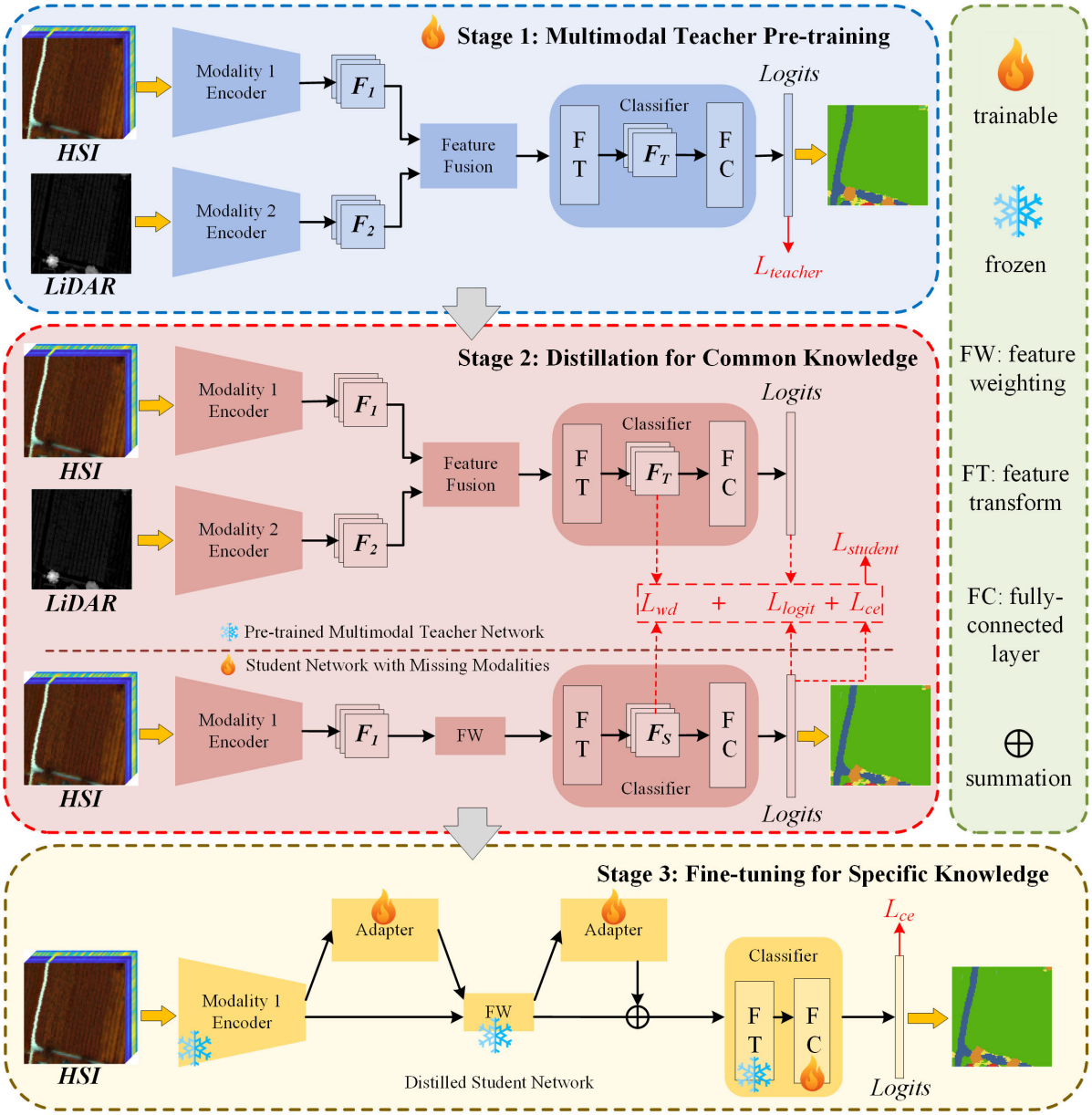


Fig. 2. Overall architecture of the proposed CSKLNet for multimodal remote sensing image classification with missing modalities. Taking the HSI–LiDAR Trento dataset as an example, F_1 and F_2 denote the extracted HSI and LiDAR features, respectively. F_T represents the complete multimodal features from the teacher network, while F_S denotes the final classification features of the student network under missing modalities. Feature transform refers to the dimensional and spatial transformation of the feature map before classification.

expressiveness of the feature. To control computational cost, a group convolution structure is used

$$X_{\text{dilate}}^{64\times S/3\times H\times W} = \text{Conv}_{3\times 1\times 1}\left(X_{\text{BN}}, \text{dilation} = (3,1,1)\right) \quad (2)$$

$$X_{\text{large}}^{256\times H\times W} = \text{Conv}_{S/3\times 1\times 1}\left(X_{\text{dilate}}, \text{group} = 2\right). \quad (3)$$

After the above steps, the original spectral information has been fully encoded into the 256-D channel representation. In order to spatially augment the output spectral tensor and achieve full synergy between spectral and spatial features, a 2-D convolution with kernel size of 3 × 3 is applied to the result obtained after processing by two 3-D convolutions, while compressing the channel dimension from 256 to 128 for subsequent use

$$X_{\text{spespa}}^{128\times H\times W} = \text{Conv}_{3\times 3}\left(X_{\text{large}}, C_{\text{in}} = 256, C_{\text{out}} = 128\right). \quad (4)$$

Through this collaborative approach, key information from both the spectral and spatial dimensions can be extracted simultaneously, achieving a more discriminative spectral–spatial feature representation.

2) Feature Weighting: Since the student model can only rely on incomplete information under missing modalities, its feature distribution deviates significantly from the multimodal teacher, potentially introducing noise and weakening the distillation process. Therefore, inspired by channel attention mechanism [56], we incorporate an FW module between the student's encoder and classifier. This module dynamically

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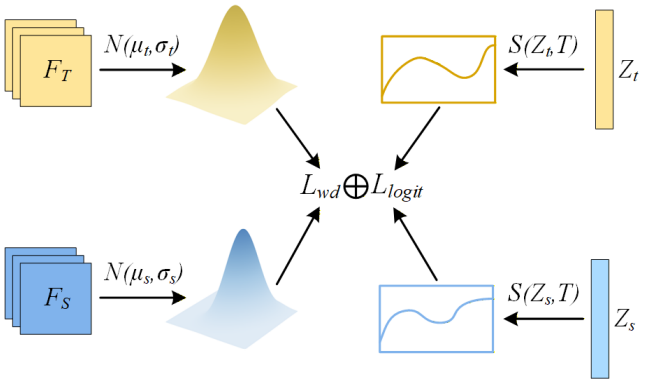


Fig. 3. Illustration of the proposed DAD method. F_T and F_S represent the teacher's and student's feature maps, respectively. $N(\mu_t, \sigma_t)$ and $N(\mu_s, \sigma_s)$ denote the Gaussian distribution modeling. Z_t and Z_s represent the teacher's and student's output logits, respectively. $S(Z_t, T)$ and $S(Z_s, T)$ denote the soften logits.

estimates the importance of each channel based on the statistical property of the input feature, enhancing reliable channels and suppressing unstable ones. In this way, the impact induced by representation distribution discrepancy can be partially reduced, thereby enhancing the stability of knowledge transfer.

Specifically, for a given input feature $F \in \mathbb{R}^{B\times C\times H\times W}$, first, global average pooling (GAP) is used

$$F_{\text{GAP}} = \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W} F_{b,c,i,j}. \tag{5}$$

The spatial information of each channel is compressed into a single statistic to construct the channel-level global receptive field. Subsequently, a fully connected network consisting of two 1×1 convolutional layers is used to compute the channel attention weights

$$\boldsymbol{s} = Sigmoid\left(\mathbf{W}_2 \cdot ReLU\left(\mathbf{W}_1 \cdot F_{\text{GAP}}\right)\right) \tag{6}$$

where $\mathbf{W}_1$ is weight matrix of the first 1×1 convolution, with dimension $\mathbb{R}^{C/R\times C}$. $\mathbf{W}_2$ is weight matrix of the second 1×1 convolution, with dimension $\mathbb{R}^{C\times C/R}$. $\boldsymbol{s}$ is final channel weight (dimension C).

The weight reflects the networks estimated importance of each channel. Finally, channel elementwise multiplication is applied to impose these weights on the input feature, achieving channel-level weighting

$$F_{\text{FW}} = \boldsymbol{s} \odot F \tag{7}$$

where $\odot$ represents the channel elementwise multiplication.

C. Distribution Alignment Distillation

In missing-modality scenarios, the inconsistency between teacher and student inputs leads to substantial representation discrepancies, which hinders effective knowledge transfer during distillation. To address this issue, we propose a DAD method, as shown in Fig. 3. Wasserstein distance-based distribution alignment and prediction-level constraints are jointly employed to alleviate representation discrepancies and facilitate effective knowledge transfer.

1) Intermediate Feature Distillation: Most traditional methods directly match the feature maps of teacher and student. However, due to the heterogeneity between teacher and student inputs, directly mimicking the teachers feature maps becomes challenging for the student model. Directly matching feature values not only fails to accurately transfer effective knowledge but also easily leads to training instability and overfitting, thereby limiting the performance improvement of the student model. To address these issues and improve the effectiveness of knowledge transfer between intermediate representations, we propose modeling features using a Gaussian distribution, minimizing the Wasserstein distance to reduce the discrepancy between teacher and student feature distributions. The details are as follows.

Given a complete multimodal input $M_i = (M_i^1, \ldots, M_i^n)$, n is the number of complete modalities. The teacher model uses (8) to extract multimodal fusion feature, which can come from any layer of the deep neural network. Here, we select the feature from the last layer before the fully connected layer for final classification

$$F_t = f_t\left(M_i^1, \ldots, M_i^n\right). \tag{8}$$

Except for replacing the input with a missing modalities combination $M_i = (M_i^1, \ldots, M_i^k)$, k is the number of available modalities, and $k < n$, the other processing of the student is similar to the teacher, using the following equation to extract feature:

$$F_s = f_s\left(M_i^1, \ldots, M_i^k\right). \tag{9}$$

Then, we obtain the intermediate feature representations F_t and F_s, with the shape of $C \times H \times W$, where C is the number of channels, H is the height of the feature map, and W is the width of the feature map. Subsequently, we adopt a parameterized distribution modeling strategy, which is more stable and easier to analyze in high-dimensional space. Specifically, we select the Gaussian distribution, the most commonly used and analytically tractable parameterized model. By modeling with the Gaussian distribution, we can approximate the latent distribution of features using a small number of parameters such as the mean and covariance, enabling efficient estimation and robust learning even in high-dimensional feature space.

First, we reshape the feature maps F to a matrix $\mathbf{F} \in \mathbb{R}^{c\times l}$ where $l = h \times w$, the ith column $f_i \in \mathbb{R}^c$ indicates a spatial feature. For f_t and f_s, we use the following equations to estimate their first moment and second moment:

$$\mu = \frac{1}{l}\sum_i f_i \tag{10}$$

$$\sigma = \frac{1}{l}\sum_i (f_i - \mu)(f_i - \mu)^T. \tag{11}$$

Based on the mean μ and covariance σ computed above, we model the distribution of the feature F in the form of a parameterized Gaussian distribution, as follows:

$$\mathcal{N}(\mu, \sigma) = \frac{1}{|2\pi\sigma|^{\frac{1}{2}}}\exp\left(-\frac{1}{2}(f-\mu)^T\sigma^{-1}(f-\mu)\right). \tag{12}$$

In this way, we can calculate the Gaussian distribution of teacher and student features. Wasserstein distance is a well-known metric used to measure the distance between two

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distributions, which preserves the geometric structure of the distributions and provides stable gradients for optimization [57]. This makes it particularly suitable for aligning heterogeneous feature distributions. Here, we let $\omega = \mathcal{N}(\mu_t, \sigma_t)$, $\nu = \mathcal{N}(\mu_s, \sigma_s)$ represent the Gaussian distributions of the teacher and student, respectively. According to the optimal transport theory [57], [58], the Wasserstein distance is defined as

$$D_{\text{wd}}(\omega, \nu) = \inf_{\pi \in \Pi(\omega,\nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|^2 \, d\pi(x, y). \tag{13}$$

Here, $\Pi(\omega, \nu)$ is the set of all joint probability distributions π whose marginal distributions are ω and ν, respectively. By expanding the objective function and leveraging the fixed marginal distributions of $X \sim \omega$ and $Y \sim \nu$, $\mathbb{E}[|X|^2]$ and $\mathbb{E}[|Y|^2]$ are constants. Thus, minimizing D_{wd} is equivalent to maximizing the cross-term expectation $\mathbb{E}_\pi[\langle X, Y \rangle]$. Due to the translation invariance property, the distance can be decomposed into contributions from the mean and the zero-mean covariance

$$D_{\text{wd}}(\omega, \nu) = \|\mu_t - \mu_s\|^2 + D_{\text{wd}}(\mathcal{N}(0, \sigma_t), \mathcal{N}(0, \sigma_s)). \tag{14}$$

For the zero-mean Gaussian distributions X_0 and Y_0, minimizing D_{wd} reduces to finding the optimal coupling π_0 that maximizes the trace of the cross-covariance matrix $\text{Tr}(\sigma_{X_0 Y_0})$. For Gaussian measures, the Optimal Transport Map $\mathbf{T}$ is linear, $Y_0 = \mathbf{T}X_0$. This matrix $\mathbf{T}$ must be the unique symmetric positive definite matrix satisfying the forward condition: $\mathbf{T}\sigma_t\mathbf{T} = \sigma_s$. Substituting $\mathbf{T}$ into the maximum term yields $\text{Tr}(\sigma_{X_0 Y_0}) = \text{Tr}(\mathbf{T}\sigma_t)$.

The unique symmetric positive definite solution $\mathbf{T}$ to the matrix equation $\mathbf{T}\sigma_t\mathbf{T} = \sigma_s$ is

$$\mathbf{T} = \sigma_t^{-1/2} \left(\sigma_t^{1/2} \sigma_s \sigma_t^{1/2} \right)^{1/2} \sigma_t^{-1/2}. \tag{15}$$

Substituting $\mathbf{T}$ into $\text{Tr}(\mathbf{T}\sigma_t)$ and utilizing the cyclic property of the Trace, the maximal cross correlation term simplifies to $\text{Tr}((\sigma_t^{1/2}\sigma_s\sigma_t^{1/2})^{1/2})$. Substituting this back into (14) decomposition establishes the final formula

$$D_{\text{wd}}(\omega, \nu) = \|\mu_t - \mu_s\|^2 + \text{Tr}\left(\sigma_t + \sigma_s - 2\left(\sigma_t^{1/2} \sigma_s \sigma_t^{1/2} \right)^{1/2} \right). \tag{16}$$

To balance the shift (mean) and the geometry (covariance) of the distribution, we add a coefficient γ before the mean term in (16). Meanwhile, to enhance the expressive power of the feature, we use a spatial pyramid pooling strategy [59] to refine the features, dividing them into $k \times k$ cells, and calculating the Wasserstein distance between teacher and student within each cell. We obtain the final loss item as follows:

$$\mathcal{L}_{\text{wd}} = \frac{1}{N_{\text{cell}}} \sum_{k=1}^{N_{\text{cell}}} \left(\gamma \left\| \mu_{t,k} - \mu_{s,k} \right\|^2 + \text{Tr}\left(\sigma_{t,k} + \sigma_{s,k} - 2\left(\sigma_{t,k}^{1/2} \sigma_{s,k} \sigma_{t,k}^{1/2} \right)^{1/2} \right) \right) \tag{17}$$

where N_{cell} is the total number of $k \times k$ cells.

2) Response Logits Distillation: The teacher model's output logits contain rich dark knowledge. These logits not only reflect the teacher model's confidence in the correct category but also reveal its discriminative relationships among nontarget categories, thereby providing information that goes beyond what hard labels can convey. Following the pioneering work of Hinton et al. [24], by minimizing the KL divergence between the teacher's and student's softened logits, students can effectively acquire the teacher's rich dark knowledge. Therefore, in addition to utilizing intermediate features for knowledge transfer, we further leverage logits to facilitate students' learning of the teacher's knowledge.

We introduce the temperature parameter T to smooth the logits, generating the teacher soft probability p and the student soft probability q

$$p_i = \frac{\exp(z_{t,i}/T)}{\sum_{j=1}^{C} \exp(z_{t,j}/T)}, \quad q_i = \frac{\exp(z_{s,i}/T)}{\sum_{j=1}^{C} \exp(z_{s,j}/T)}. \tag{18}$$

Here, z_t and z_s are the C-dimensional logits vectors, representing C categories, and $T \in \mathbb{R}^+$ is the temperature parameter used to control the smoothness of the distribution.

The distillation loss $\mathcal{L}_{\text{logit}}$ is defined as the KL divergence $D_{\text{KL}}(p\|q)$ of the student distribution q with respect to the teacher distribution p, multiplied by the gradient scaling factor T^2

$$\mathcal{L}_{\text{logit}} = T^2 \cdot D_{\text{KL}}(p\|q) = T^2 \sum_{i=1}^{C} p_i \log\left(\frac{p_i}{q_i} \right). \tag{19}$$

This loss function drives the student network to learn the dark knowledge of the teacher network by minimizing $-\sum_i p_i \log q_i$. The multiplication by T^2 is intended to counteract the gradient magnitude decay effect observed at high T value, ensuring consistency with the gradient scale of the standard classification loss.

D. Adapter Fine-Tuning

Through distillation, the student model inherits multimodal knowledge from the teacher model. However, the teacher primarily learns a shared joint representation during multimodal feature fusion, which captures high-level semantic information but may overlook fine-grained modality-specific characteristics. Consequently, directly deploying the distilled student model under missing modalities may limit its ability to exploit the discriminative information of the available modality. To address this issue, we further fine-tune the distilled student model to readapt its representation to the available modality, thereby enhancing the learning of modality-specific information and improving classification performance.

To help the student model better adapt information from the available modalities, we introduce lightweight, trainable adapter modules at two key locations: one is placed between the feature extractor and the FW module, and the other between the FW module and the classifier. The adapter works as follows:

$$F_{\text{adapted}} = \text{BN}\left(\mathbf{W}_2 \otimes \left(\delta\left(\text{BN}\left(\mathbf{W}_1 \otimes F_{\text{original}} \right) \right) \right) \right) \tag{20}$$

where F_{original} is the input feature, F_{adapted} is the adapted feature, both with c channels, $\mathbf{W}_1$ is the first convolutional

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kernel, reducing channels from c to c/r, $\mathbf{W}_2$ is the second convolutional kernel, restoring channels from c/r to c. δ and BN represent the ReLU and batch normalization operation, respectively. $\otimes$ denotes the convolution operation.

During the fine-tuning phase, only the parameters of these adapters and the fully connected layer in the classifier are updated. This enables the student model to adapt to the available modality while preserving the multimodal knowledge acquired through distillation. The fine-tuning process is illustrated as follows. First, the frozen feature extractor extracts an initial representation from the available modality. The resulting feature is then processed through two parallel streams. In the adaptation stream, it is sequentially passed through the two adapters to progressively refine the representation. In parallel, the main stream feeds the feature into the frozen FW module to retain the original knowledge representation. Then, the outputs of the two streams are subsequently combined through elementwise addition to obtain the refined feature representation, which is then fed into the classifier for final prediction. The trainable components in this fine-tuning stage are optimized using the cross-entropy loss, as shown in the following equation:

$$\begin{aligned} F_{\text{adapted}} &= \mathrm{A}_2\left(\mathrm{A}_1\left(F_{\text{original}}\right)\right) \\ F_{\text{fused}} &= F_{\text{original}} + F_{\text{adapted}} \\ L &= Classifier\left(F_{\text{fused}}\right) \end{aligned} \tag{21}$$

$$\mathcal{L}_{\text{ce}} = -\sum_{C=1}^{C} Y_C \log\left(\frac{\exp\left(L_C\right)}{\sum_{j=1}^{C}\exp\left(L_j\right)}\right) \tag{22}$$

where A represents the adapter, L is the output logits, Y is the ground truth (GT), and C is the class number.

Through the above process, the student can further utilize the available modality to learn modality-specific information.

E. Training Paradigm

Our proposed method's training consists of three parts: teacher model training, student model distillation learning, and AFT learning.

1) Teacher Model Training: The teacher model is trained on complete multimodal data to learn comprehensive multimodal representations. Specifically, the hyperspectral encoder is used for HSI feature extraction, while the remaining modalities employ the 2-D convolution encoder from MDL_RS. Other configurations follow Section III-B. The teacher model is optimized using the cross-entropy loss

$$\mathcal{L}_{\text{teacher}} = -\sum_{c=1}^{c} Y_c \log\left(\frac{\exp\left(L_{t,c}\right)}{\sum_{j=1}^{c}\exp\left(L_{t,j}\right)}\right) \tag{23}$$

where L_t is the teacher's output logits, c is the class number, and Y represents the GT.

2) Student Model Distillation Learning: In this study, the student model operates on a single modality. Therefore, to minimize network architecture difference, the same modality encoder and classifier are adopted, and an FW module between them (as described in Section III-B). The optimization objective consists of classification loss and distillation loss, including intermediate feature and output logits distillation

$$\mathcal{L}_{\text{student}} = \mathcal{L}_{\text{ce}} + \alpha\mathcal{L}_{\text{wd}} + \beta\mathcal{L}_{\text{logit}}. \tag{24}$$

Here, $\mathcal{L}_{\text{ce}}$ is cross-entropy loss same as (23), $\mathcal{L}_{\text{wd}}$ and $\mathcal{L}_{\text{logit}}$ are set to (17) and (19), respectively. α and β are the weighting factors used to balance the loss components.

3) AFT Learning: In this stage, most parameters of the distilled student model are frozen to preserve the knowledge acquired during distillation, while only the adapters and the fully connected layer are updated. The optimization is performed using (22).

IV. Experiments

A. Datasets

To evaluate the effectiveness of the proposed method, we selected three widely used multimodal land cover classification datasets: HSI–LiDAR Houston2013, HSI–LiDAR Trento, and HSI–MS Houston2013, and constructed a multimodal dataset for mangrove species classification: HSI–MS QAI-Mangrove. A brief description of these datasets is provided below.

1) HSI–LiDAR Houston2013: This is a multimodal dataset containing both HSI and LiDAR images. The HSI data comprises 144 bands ranging from 380 to 1050 nm, while the LiDAR data is a single-band grayscale image with the same spatial resolution as the HSI (2.5 m). The data collection area covered the University of Houston campus and the surrounding urban area. In total, the dataset contains 15 029 samples across 15 land cover categories, of which 2832 are used for training and 12 197 for testing, as detailed in Table I.

2) HSI–LiDAR Trento: The second dataset, which similarly consists of one HSI with 63 bands and one LiDAR image. Specifically, this scene consists of 166 $\times$ 600 pixels with a spatial resolution of 1 m. The data were collected from a rural area of the city of Trento in southern Italy. In total, the dataset contains 30 414 samples across six land cover categories, of which 819 are used for training and 29 595 for testing, as detailed in Table I.

3) HSI–MS Houston2013: To make full use of the high spectral information of the HSI and high spatial information of the MS, Hong et al. [9] generated the HSI–MS Houston2013 dataset from the HSI data in the IEEE GRSS data fusion contest 2013 by degrading the spatial and spectral domains. The MS data comprises eight bands with the same size as the original HSI. The data collection area covered the University of Houston campus and the surrounding urban area. In total, the dataset contains 15 029 samples across 15 land cover categories, of which 2832 are used for training and 12 197 for testing, as detailed in Table I.

4) HSI–MS QAI-Mangrove: This is a multimodal mangrove species classification dataset, containing HSI and MS images of the Qi'ao Island Mangrove Nature Reserve in Guangdong, China. Specifically, the HSI data were collected from the GF-5B satellite, from which 296 spectral bands were retained after preprocessing, with a spatial resolution of 30 m. For

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TABLE I

NUMBER OF TRAIN AND TEST SAMPLES FOR EACH CLASS OF EXPERIMENTAL DATASETS

No.	Houston2013				Trento				QAI-Mangrove			
	Class Name	Train	Test	Total	Class Name	Train	Test	Total	Class Name	Train	Test	Total
1	Healthy grass	198	1053	1251	Apple trees	129	3905	4034	Kandelia obovata	16	80	96
2	Stressed grass	190	1064	1254	Buildings	125	2778	2903	Sonneratia apetala	32	592	624
3	Synthetic grass	192	505	697	Ground	105	374	479	Aegiceras corniculatum	16	32	48
4	Tree	188	1056	1244	Wood	154	8969	9123	Acanthus ilicifolius	16	96	112
5	Soil	186	1056	1242	Vineyard	184	10317	10501	Water area	16	192	208
6	Water	182	143	325	Roads	122	3252	3374	Acrostichum aureum	16	112	128
7	Residential	196	1072	1268	-	-	-	-	Spartina alterniflora	16	80	96
8	Commercial	191	1053	1244	-	-	-	-	-	-	-	-
9	Road	193	1059	1252	-	-	-	-	-	-	-	-
10	Highway	191	1036	1227	-	-	-	-	-	-	-	-
11	Railway	181	1054	1235	-	-	-	-	-	-	-	-
12	Parking lot 1	192	1041	1233	-	-	-	-	-	-	-	-
13	Parking lot 2	184	285	469	-	-	-	-	-	-	-	-
14	Tennis court	181	247	428	-	-	-	-	-	-	-	-
15	Running track	187	473	660	-	-	-	-	-	-	-	-
Total	-	2832	12197	15029	-	819	29595	30414	-	128	1184	1312

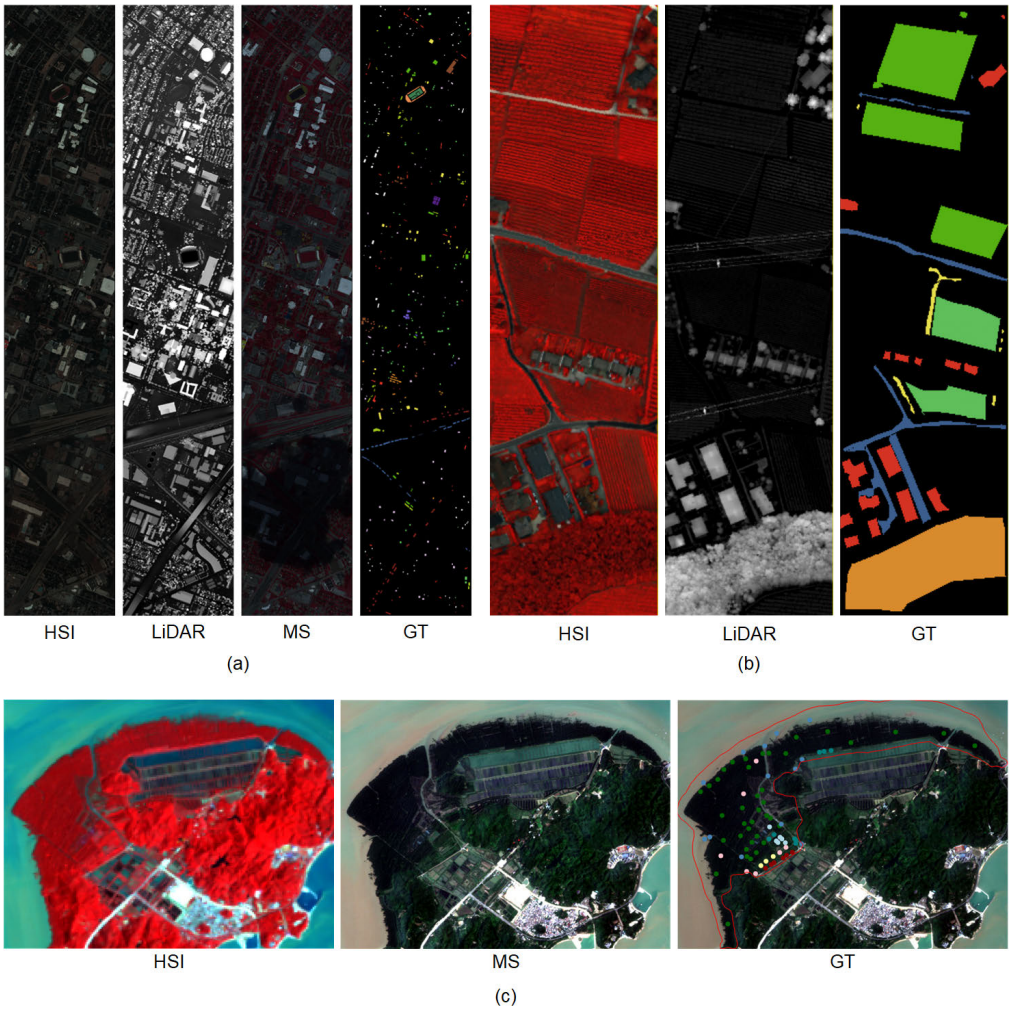


Fig. 4. Multimodal datasets used in this work. (a) HSI–LiDAR/HSI–MS Houston2013. (b) HSI–LiDAR Trento. (c) HSI–MS QAI-Mangrove. "GT" is "ground truth."

the MS data, we adopted the Sentinel-2 image, preserving 12 bands and resampling them to a unified spatial resolution of 10 m. Overall, the dataset contains seven classes and a total of 1312 labeled samples, among which 128 samples are used for training and 1184 for testing, as shown in Table I.

The visualized images and GT maps corresponding to the four datasets are presented in Fig. 4.

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B. Experimental Setup

1) Evaluation Metrics: In this work, we adopt a multimodal land cover classification task to evaluate the effectiveness of the proposed method, as it serves as a representative and challenging application for multimodal learning. To ensure a thorough and reliable assessment, we employ three widely used evaluation metrics in land cover classification: overall accuracy (OA), average accuracy (AA), and the kappa coefficient (κ).

2) Implementation Details: All experiments were implemented under the PyTorch framework using Python 3.8 and conducted on a computing platform equipped with an NVIDIA GeForce RTX 3090 GPU (24-GB memory). Model training was performed using the stochastic gradient descent (SGD) optimizer with a momentum of 0.9, and the batch size was fixed at 32. Regarding the training schedule, both the teacher model training stage and the student distillation stage were trained for 500 epochs on all four datasets. During the AFT stage, the number of training epochs was set to 500 for the QAI dataset, while the remaining three datasets were trained for 50 epochs. For the initial learning rate, $1e^{-4}$ was adopted for the QAI dataset across all three stages, whereas $1e^{-3}$ was used for the other three datasets. Throughout training, the classical cosine learning rate policy was employed to dynamically adjust the learning rate. In addition, the input patch size was uniformly set to 7×7 for all experiments.

3) Comparison Methods: To evaluate the effectiveness of the proposed CSKLNet in the land cover classification task under missing modalities, we compare it with two categories of representative approaches designed to deal with missing modalities. The first category is the unified model method, which train a single model to accommodate randomly missing modalities. For this category, we select three state-of-the-art (SOTA) methods, DGDNet [26], ShaSpec [20] and mmFormer [19]. The second category is nonunified model method, which train dedicated models for each specific missing scenario, thereby enabling more targeted handling of missing modalities. In this category, we select four SOTA methods, MSHNet [31], CLKD [25], ProtoKD [29] and PMKL [30].

To ensure fairness in the comparison, all comparison methods used MDL_RS as the base model architecture in the experiments. Apart from the specific modules and parameter settings required by each method, all other training configurations remained consistent.

C. Results and Analysis on HSI–LiDAR Houston2013

First, we conducted a series of baseline experiments, including "H-L-Net," "H-Net," and "L-Net," to investigate the impact of missing modalities on conventional multimodal models that lack dedicated mechanisms for handling missing modalities. All three methods adopt exactly the same network architecture and parameter configuration as the teacher used in our proposed approach, and are trained using the complete HSI+LiDAR modality combination. During the testing phase, different modality input strategies are employed. Specifically, "H-L-Net" performs inference using both HSI and LiDAR, while "H-Net" and "L-Net" rely on HSI or LiDAR, respectively. The absent modality is replaced with a zero-valued tensor of the same size. As reported in Table II, when LiDAR or HSI is missing during inference, the OA of "H-Net" and "L-Net" decreases by 16.36% and 74.52%, respectively, compared with the full modalities setting of "H-L-Net." This substantial performance degradation indicates that conventional multimodal models heavily depend on the synergistic representation learned from multiple modalities during training. Once a modality is missing at the inference stage, the resulting feature distribution becomes severely mismatched with that observed during training, leading to a dramatic decline in model performance.

In contrast, CSKLNet (Ours) demonstrates strong robustness with missing modalities. Whether HSI or LiDAR is absent, our approach consistently outperforms the baseline single-modality testing across all evaluation metrics. This improvement benefits from the two-stage learning strategy of the student, where knowledge distillation transfers multimodal common information from the teacher, and the AFT further enhances the exploitation of modality-specific information from the remaining modalities, thereby alleviating the information deficiency caused by missing modalities. As shown in Table II, DAD + AF consistently outperforms DAD alone under two missing modalities conditions.

TABLE II

PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE HSI–LIDAR HOUSTON2013 DATASET UNDER MISSING HSI OR LIDAR MODALITY. BOLD AND UNDERLINED VALUES INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Method	Training	Testing	OA (%)	AA (%)	$\kappa \times 100$
Baseline					
H-L-Net[1]	HSI+LiDAR	HSI+LiDAR	91.91	92.65	91.21
H-Net[2]	HSI+LiDAR	HSI	70.21	75.55	67.69
L-Net[3]	HSI+LiDAR	LiDAR	17.39	13.27	9.50
W/o[6]LiDAR modality					
DGDNet	HSI+LiDAR	HSI	84.47	86.58	83.21
mmFormer	HSI+LiDAR	HSI	84.96	86.84	83.74
ShaSpec	HSI+LiDAR	HSI	81.34	83.40	79.80
MSHNet	HSI+LiDAR	HSI	85.51	87.33	84.32
ProtoKD	HSI+LiDAR	HSI	86.36	87.84	85.22
PMKL	HSI+LiDAR	HSI	85.02	87.11	83.81
CLKD	HSI+LiDAR	HSI	84.24	86.11	82.90
Ours (DAD)[4]	HSI+LiDAR	HSI	<u>90.15</u>	<u>91.04</u>	<u>89.30</u>
Ours (DAD+AF)[5]	HSI+LiDAR	HSI	**90.63**	**91.42**	**89.82**
W/o[6]HSI modality					
DGDNet	HSI+LiDAR	LiDAR	59.26	61.09	56.02
mmFormer	HSI+LiDAR	LiDAR	56.06	57.49	52.51
ShaSpec	HSI+LiDAR	LiDAR	48.81	46.73	44.39
MSHNet	HSI+LiDAR	LiDAR	60.60	62.99	57.43
ProtoKD	HSI+LiDAR	LiDAR	60.81	63.02	57.69
PMKL	HSI+LiDAR	LiDAR	59.91	61.56	56.73
CLKD	HSI+LiDAR	LiDAR	55.06	56.07	51.54
Ours (DAD)[4]	HSI+LiDAR	LiDAR	<u>61.97</u>	<u>64.14</u>	<u>58.90</u>
Ours (DAD+AF)[5]	HSI+LiDAR	LiDAR	**62.15**	**64.38**	**59.12**

[1] "H-L-Net" refers to conventional multimodal model that uses HSI and LiDAR for both training and inference.
[2] "H-Net" refers to conventional multimodal model that uses HSI and LiDAR for training, HSI for inference.
[3] "L-Net" refers to conventional multimodal model that uses HSI and LiDAR for training, LiDAR for inference.
[4] "DAD" denotes Distribution Alignment Distillation.
[5] "AF" denotes Adapter Fine-Tuning.
[6] "W/o" indicates the modality missing during testing.

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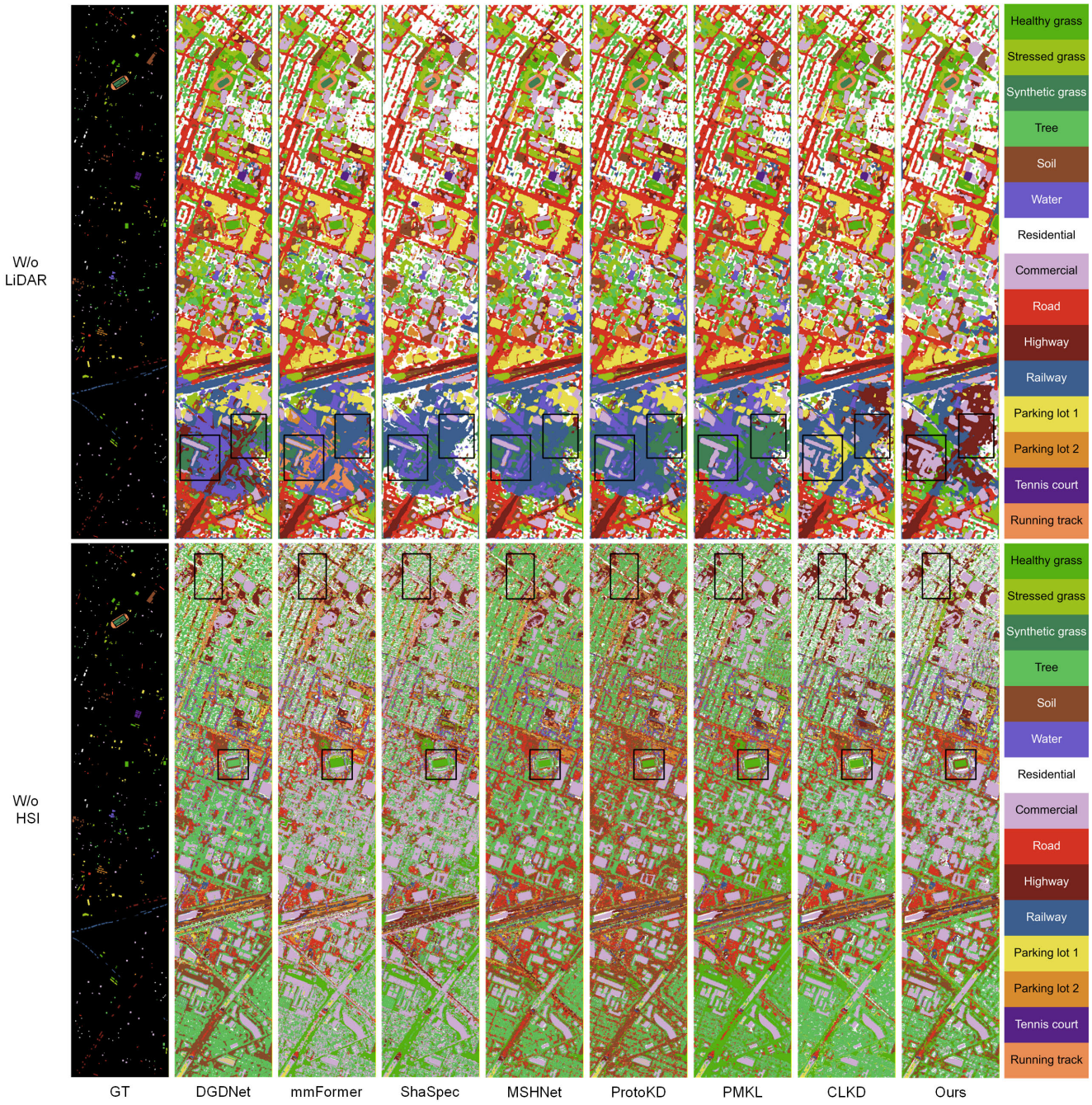


Fig. 5. Classification maps of different compared methods with missing LiDAR or HSI modality on HSI–LiDAR Houston2013 dataset.

In terms of OA, DAD + AF achieves an improvement of 0.18% and 0.48% when HSI or LiDAR is missing, respectively, confirming the effectiveness of the fine-tuning stage for reinforcing modality-specific feature learning. Moreover, under both missing-modality scenarios, CSKLNet (Ours) consistently outperforms seven SOTA methods across all evaluation metrics. Specifically, our method surpasses DGDNet, mmFormer, ShaSpec, MSHNet, ProtoKD, PMKL, CLKD by 2.89%, 6.09%, 13.34%, 1.55%, 1.34%, 2.24%, and 7.09% in the OA when HSI is missing, and 6.16%, 5.67%, 9.29%, 5.12%, 4.27%, 5.61%, and 6.39% in the OA when LiDAR is missing. These results demonstrate the superior robustness and effectiveness of the proposed CSKLNet for multimodal classification with missing modalities.

Finally, classification maps of all methods under two missing-modality scenarios are presented. Results of "W/o LiDAR" in Fig. 5 show the classification maps when the LiDAR modality is missing. Taking the commercial (left black rectangle) and highway (right black rectangle) classes as examples, our method can accurately distinguish the target categories, whereas the other seven methods exhibit obvious misclassification and incomplete identification. Results of "W/o HSI" in Fig. 5 present the classification maps when the HSI modality is missing. For healthy grass (middle black

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TABLE III
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE HSI–LIDAR TRENTO DATASET UNDER MISSING HSI OR LIDAR MODALITY. BOLD AND UNDERLINED VALUES INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Method	Training	Testing	OA (%)	AA (%)	$\kappa \times 100$
Baseline					
H-L-Net[1]	HSI+LiDAR	HSI+LiDAR	99.30	98.62	99.06
H-Net[2]	HSI+LiDAR	HSI	64.67	75.47	55.66
L-Net[3]	HSI+LiDAR	LiDAR	37.23	43.95	28.43
W/o[6]LiDAR modality					
DGDNet	HSI+LiDAR	HSI	97.58	96.01	96.76
mmFormer	HSI+LiDAR	HSI	97.52	96.15	96.68
ShaSpec	HSI+LiDAR	HSI	95.87	94.71	94.48
MSHNet	HSI+LiDAR	HSI	97.99	96.84	97.31
ProtoKD	HSI+LiDAR	HSI	97.60	96.13	96.79
PMKL	HSI+LiDAR	HSI	97.86	96.11	97.13
CLKD	HSI+LiDAR	HSI	97.85	96.09	97.12
Ours (DAD)[4]	HSI+LiDAR	HSI	<u>98.31</u>	<u>97.29</u>	<u>97.73</u>
Ours (DAD+AF)[5]	HSI+LiDAR	HSI	**98.34**	**97.35**	**97.78**
W/o[6]HSI modality					
DGDNet	HSI+LiDAR	LiDAR	90.66	85.03	87.65
mmFormer	HSI+LiDAR	LiDAR	90.71	84.52	87.71
ShaSpec	HSI+LiDAR	LiDAR	90.21	83.96	86.97
MSHNet	HSI+LiDAR	LiDAR	92.70	86.33	90.30
ProtoKD	HSI+LiDAR	LiDAR	92.13	85.70	89.56
PMKL	HSI+LiDAR	LiDAR	92.18	85.78	89.62
CLKD	HSI+LiDAR	LiDAR	92.55	86.01	90.12
Ours (DAD)[4]	HSI+LiDAR	LiDAR	<u>93.20</u>	<u>86.93</u>	<u>90.99</u>
Ours (DAD+AF)[5]	HSI+LiDAR	LiDAR	**93.75**	**87.49**	**91.70**

[1] "H-L-Net" refers to conventional multimodal model that uses HSI and LiDAR for both training and inference.
[2] "H-Net" refers to conventional multimodal model that uses HSI and LiDAR for training, HSI for inference.
[3] "L-Net" refers to conventional multimodal model that uses HSI and LiDAR for training, LiDAR for inference.
[4] "DAD" denotes Distribution Alignment Distillation.
[5] "AF" denotes Adapter Fine-Tuning.
[6] "W/o" indicates the modality missing during testing.

rectangle) and residential (top black rectangle), our method yielded the most complete classification with fewer misclassifications. In contrast, DGDNet almost entirely misclassified healthy grass as tree, while ProtoKD can reasonably identify healthy grass but misclassified a large portion of residential as tree. These visualization results further demonstrate the superiority of the proposed method.

D. Results and Analysis on HSI–LiDAR Trento

From the baseline results in Table III, it can be observed that when the HSI modality is missing, the OA decreases by 62.07%, and when the LiDAR modality is missing, the OA decreases by 34.63%. These results indicate that even in a rural scene with relatively few land cover categories, conventional multimodal classification model suffers from a substantial performance degradation when encountering missing modalities during inference.

Notably, CSKLNet (Ours) is also effective in handling missing modalities with high heterogeneous modality combinations in rural scene. Compared with "H-Net" and "L-Net," our method achieves improvements of 33.67% and 56.52% in terms of OA, respectively. Furthermore, with the two different missing-modality scenarios, CSKLNet (Ours) consistently outperforms seven SOTA methods across all evaluation metrics. Specifically, our method surpasses DGDNet, mmFormer, ShaSpec, MSHNet, ProtoKD, PMKL, CLKD by 3.09%, 3.04%, 3.54%, 1.05%, 1.62%, 1.57%, and 1.20% in the OA when HSI is missing, and 0.76%, 0.82%, 2.47%, 0.35%, 0.74%, 0.48%, and 0.49% in the OA when LiDAR is missing. These quantitative comparisons demonstrate the superiority of the proposed CSKLNet. Although our learning strategy consists of two stages, the results in Table III indicate that even when only the first stage is applied, our method still outperforms all competing methods across the three evaluation metrics. This suggests that the student is able to effectively acquire the complete multimodal information from the teacher through our DAD method. The second stage AFT further provides additional gains in classification performance, highlighting the effectiveness of combining common multimodal information with modality-specific information.

Subsequently, the classification results for the two missing modalities are presented. Results of "W/o LiDAR" in Fig. 6 present the classification results under the missing LiDAR modality scenarios. As observed in the left black rectangular region, compared with the other seven methods, CSKLNet (Ours) produces more complete and coherent road extraction and yields fewer misclassifications in vineyard areas. Results of "W/o HSI" in Fig. 6 present the classification results under the missing HSI modality. As shown in the right black rectangle, the seven comparison methods misclassify vineyard as apple trees to varying degrees, whereas the proposed method exhibits noticeably fewer misclassification. The comparison of classification maps further confirms that CSKLNet (Ours) maintains stable performance when applied to missing modalities in the rural scene.

E. Results and Analysis on HSI–MS Houston2013

As the results shown in Table IV, the baseline results indicate that when only the MS modality is available for inference, the OA drops by 58.41%, and when inference relies solely on the HSI modality, the OA drops by 29.85%. These results demonstrate that even for homogeneous modality combinations with fewer intermodality differences, the absence of any modality can still lead to a substantial degradation in the performance of a conventional classification model.

Nevertheless, CSKLNet (Ours) is still able to effectively mitigate the performance degradation caused by the missing of any modality, indicating that our approach is not limited to heterogeneous modality combination but is also applicable to homogeneous ones, demonstrating favorable generalization ability. Specifically, our method achieves substantial improvements over the baseline methods "H-Net" and "M-Net" across all evaluation metrics. Moreover, when the HSI modality is missing, our method improves the OA by 2.03%, 1.50%, 13.18%, 0.62%, 0.55%, 1.03%, and 1.14% compared with DGDNet, mmFormer, ShaSpec, MSHNet, ProtoKD, PMKL, and CLKD, respectively. When the MS modality is missing, the corresponding OA improvements are 7.48%, 6.32%, 17.65%, 5.92%, 7.28%, 5.87%, and 6.47%, respectively.

Two observations can be drawn from the experimental results. First, when the MS modality is missing, our method even outperforms the full modality setting across all evaluation metrics. This phenomenon can be attributed to the fact that

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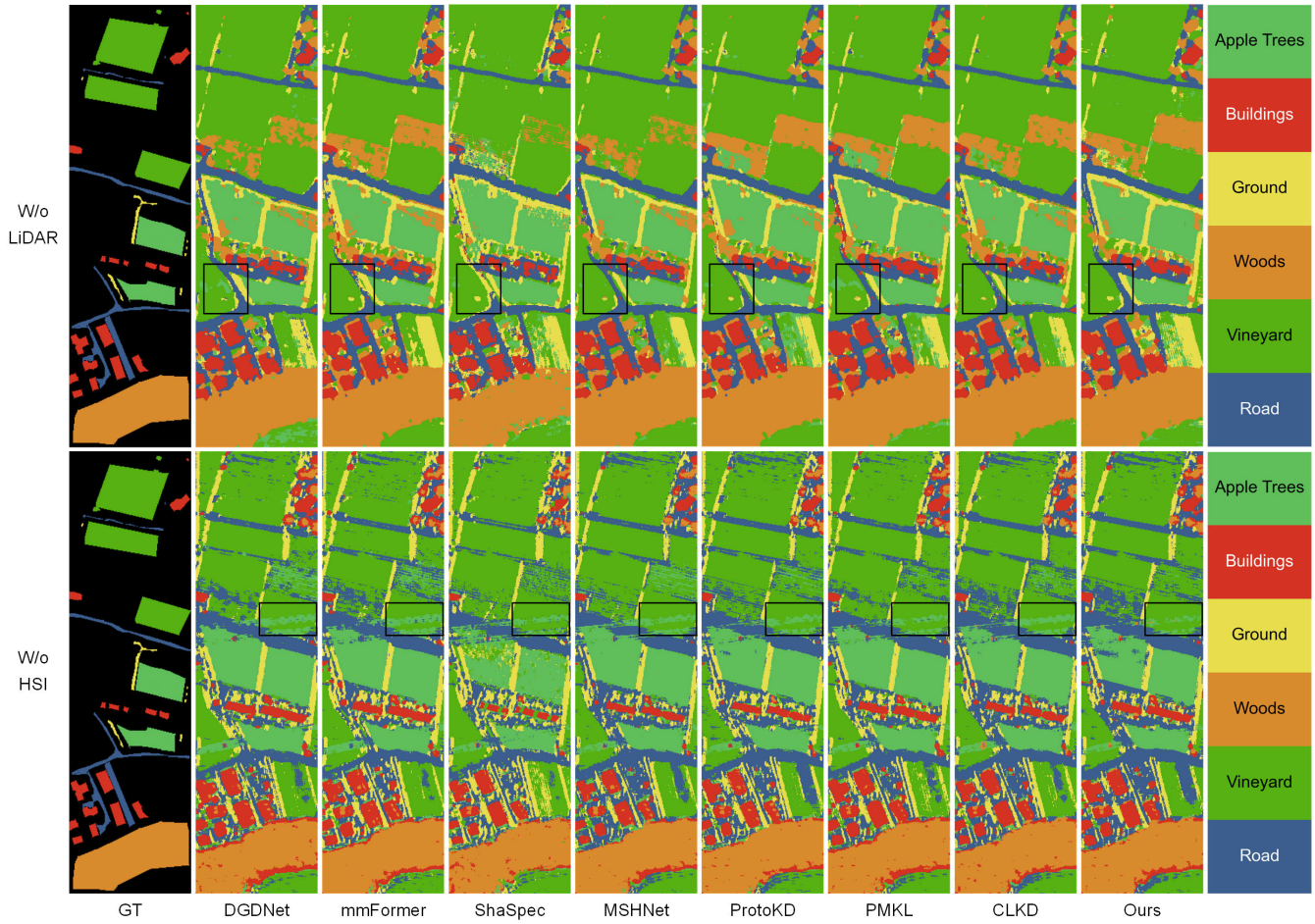


Fig. 6. Classification maps of different compared methods with missing LiDAR or HSI modality on HSI–LiDAR Trento dataset.

conventional multimodal models tend to emphasize common representations across modalities, which may inadvertently weaken the learning of discriminative features specific to individual modality. In contrast, the AFT in our framework further enhances the learning of available modality-specific information. Combined with the common knowledge distilled in the first stage, the resulting model exhibits greater discriminative capability. Second, the seven comparison methods generally achieve better performance when HSI is missing, while an opposite trend is observed for our method. This can be explained by the tailored feature extraction strategy applied to HSI in our framework. Specifically, the hyperspectral encoder effectively alleviates the spectral redundancy inherent in HSI, thereby enabling more efficient exploitation of the rich spectral information. As a result, our method benefits more substantially from the presence of HSI.

Subsequently, the classification maps are presented. Results of "W/o MS" in Fig. 7 illustrate the results when the MS modality is missing. Taking the Highway (upper black rectangle) and commercial (lower black rectangle) classes as examples, CSKLNet (Ours) successfully identifies the highway regions, whereas the other seven methods fail to recognize this class within the corresponding areas. For the commercial class, our method produces the most complete and coherent classification results, while the other methods only capture partial regions. Results of "W/o HSI" in Fig. 7 present the classification results when the HSI modality is missing. Considering the healthy grass (left black rectangle) and commercial (right black rectangle) classes, our method correctly identifies healthy grass, whereas none of the other seven methods achieves reliable recognition. Moreover, for the commercial class, CSKLNet (Ours) yields the most regular and complete classification. These visual comparisons further highlight the robustness and effectiveness of the proposed method when applied to homogeneous modality combination.

F. Results and Analysis on HSI–MS QAI-Mangrove

Similarly, based on the results in the baseline in Table V, "H-Net" and "M-Net" show a decrease in OA of 19.6% and 14.11%, respectively, compared with "H-M-Net," which once again verify the shortcomings of traditional multimodal models with missing modalities. Our proposed method still enables multimodal model to maintain considerable classification performance with missing modalities. Specifically, compared with "H-Net," our method improves OA by 18.58% when MS is missing, and compared with "M-Net," our method improves OA by 15.04% when HSI is missing. Furthermore, compared with the seven SOTA methods, when HSI is missing, our method achieves improvements in OA of 0.17%, 2.11%, 0.93%, 6.00%, 3.21%, 4.90%, and 7.18%, respectively; when

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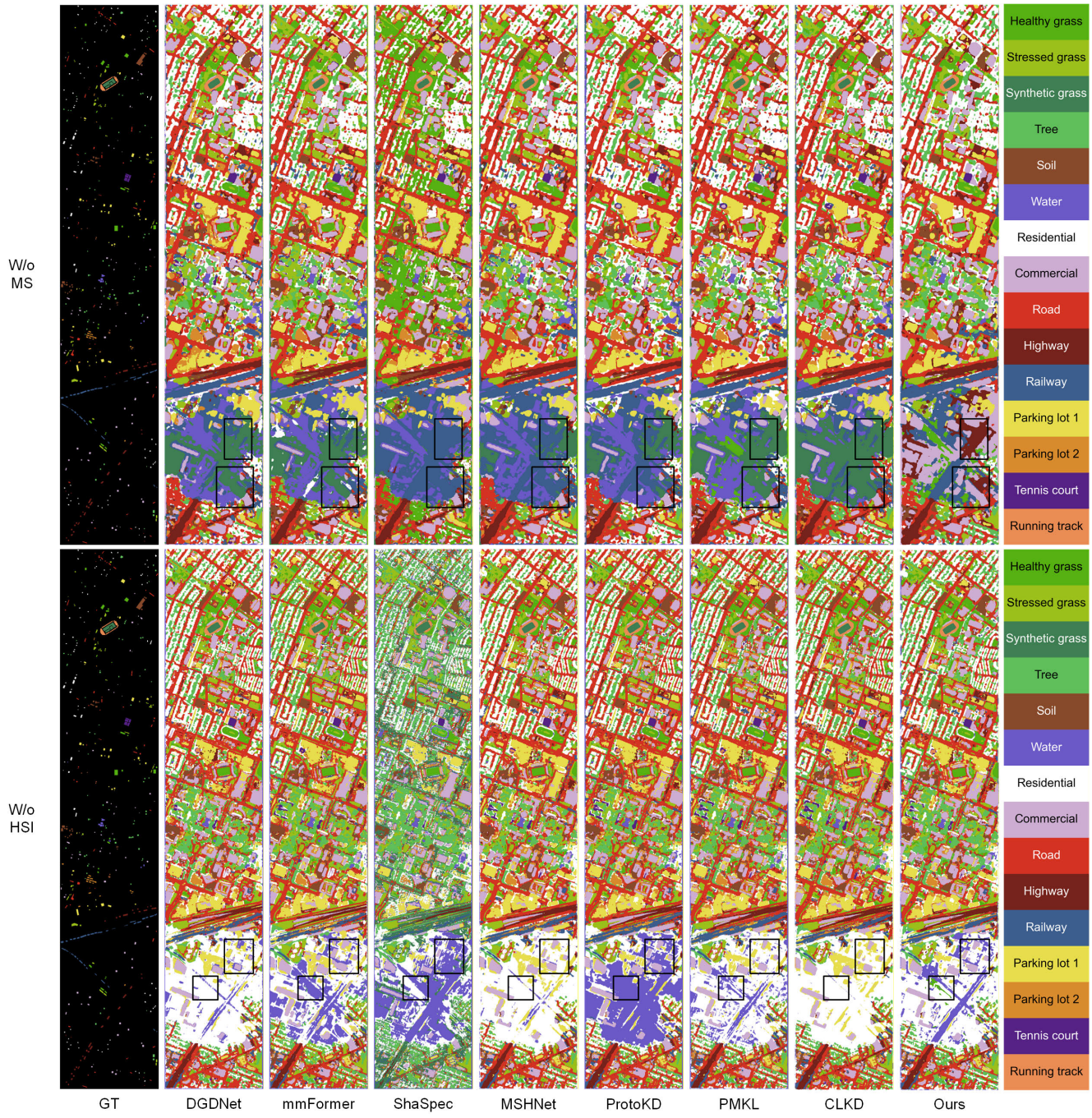


Fig. 7. Classification maps of different compared methods with missing MS or HSI modality on the HSI–MS Houston2013 dataset.

MS is missing, our method achieves improvements in OA of 9.29%, 5.49%, 9.29%, 8.28%, 11.91%, 11.40%, and 12.67%, respectively.

It is worth noting that when HSI is missing, after the second-stage learning, our method outperforms "H-M-Net" in all evaluation metrics. This may be attributed to the AFT learning effectively integrating the preserved MS-specific information with the multimodal common information learned by distillation, further improving the model's classification ability. However, when MS is missing, our second-stage learning only improves AA, while the other two metrics slightly decrease. This behavior may be related to the spectral and spatial characteristics of mangrove species. Although HSI provides rich spectral information, the spectral separability among different species is not uniform, and some species exhibit similar spectral responses. Meanwhile, the 30-m spatial resolution of HSI limits the representation of fine-grained species distributions and may introduce mixed pixels in areas where different species coexist. Consequently, AFT may enhance some species-specific features but introduce uninformative variations for spectrally similar or mixed species.

Finally, we present the classification maps under two missing-modality scenarios. Results of "W/o MS" in Fig. 8 show the classification results of different methods when MS

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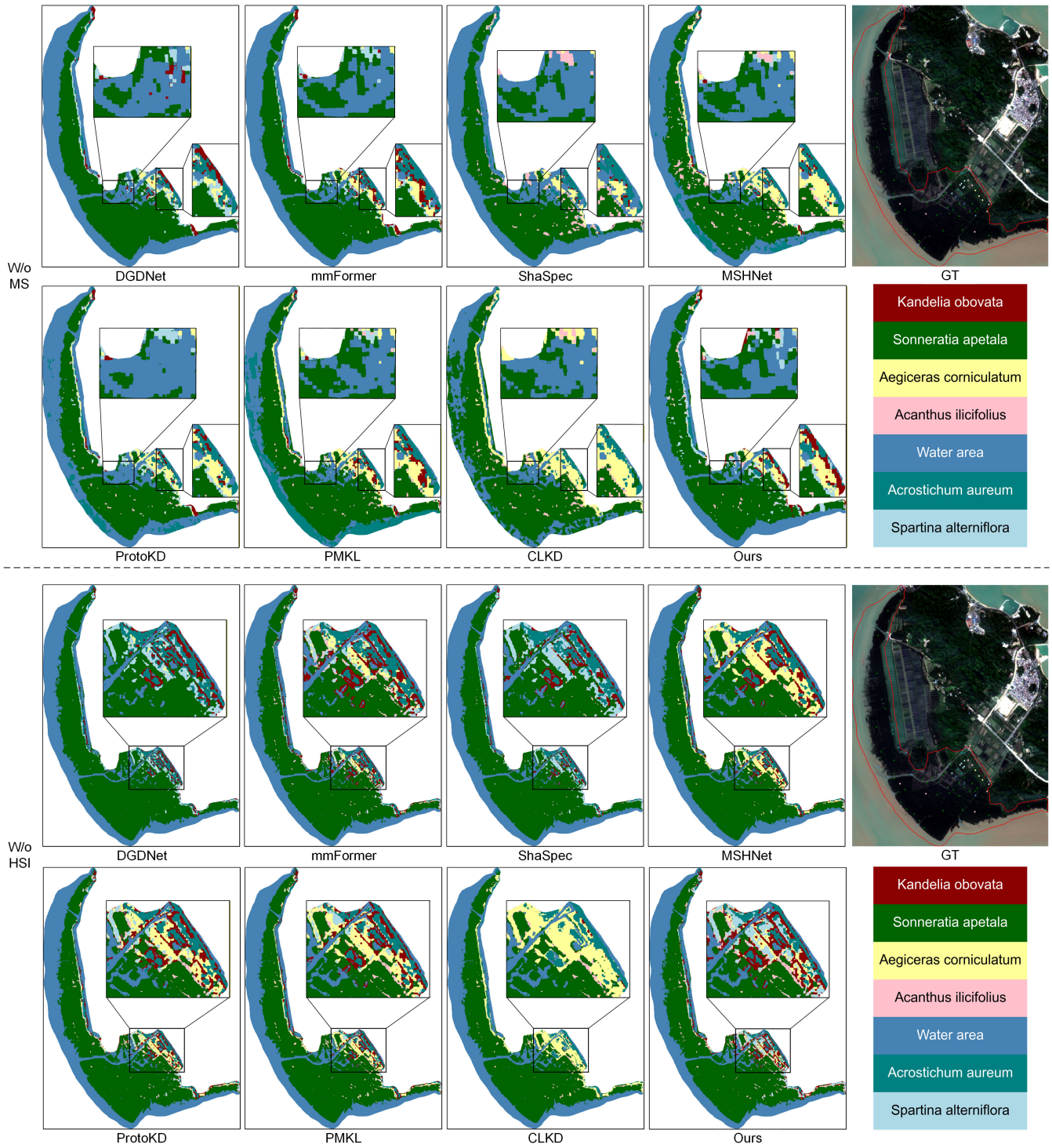


Fig. 8. Classification maps of different compared methods with missing MS or HSI modality on HSI–MS QAI-Mangrove dataset.

is missing. Taking *Sonneratia apetala* and water area as examples (left black rectangle), most methods tend to misclassify *S. apetala* as water area. Compared with our approach, the other methods show limited capability in recognizing *Kandelia obovata* and cannot even separate it from other categories. (right black rectangle). Results of "W/o HSI" in Fig. 8 present the classification maps when HSI is missing, where the black rectangle indicates the region with the highest species diversity. As observed, MSHNet, ProtoKD, PMKL, and CLKD show notable misclassifications where other classes are classified as *Aegiceras corniculatum*, DGDNet and ShaSpec can barely identify *A. corniculatum*. Although mmFormer recognizes *A. corniculatum* relatively accurately, it fails to fully identify *Spartina alterniflora*. Overall, our method achieves competitive performance in the missing-modalities classification task under wetland scenarios.

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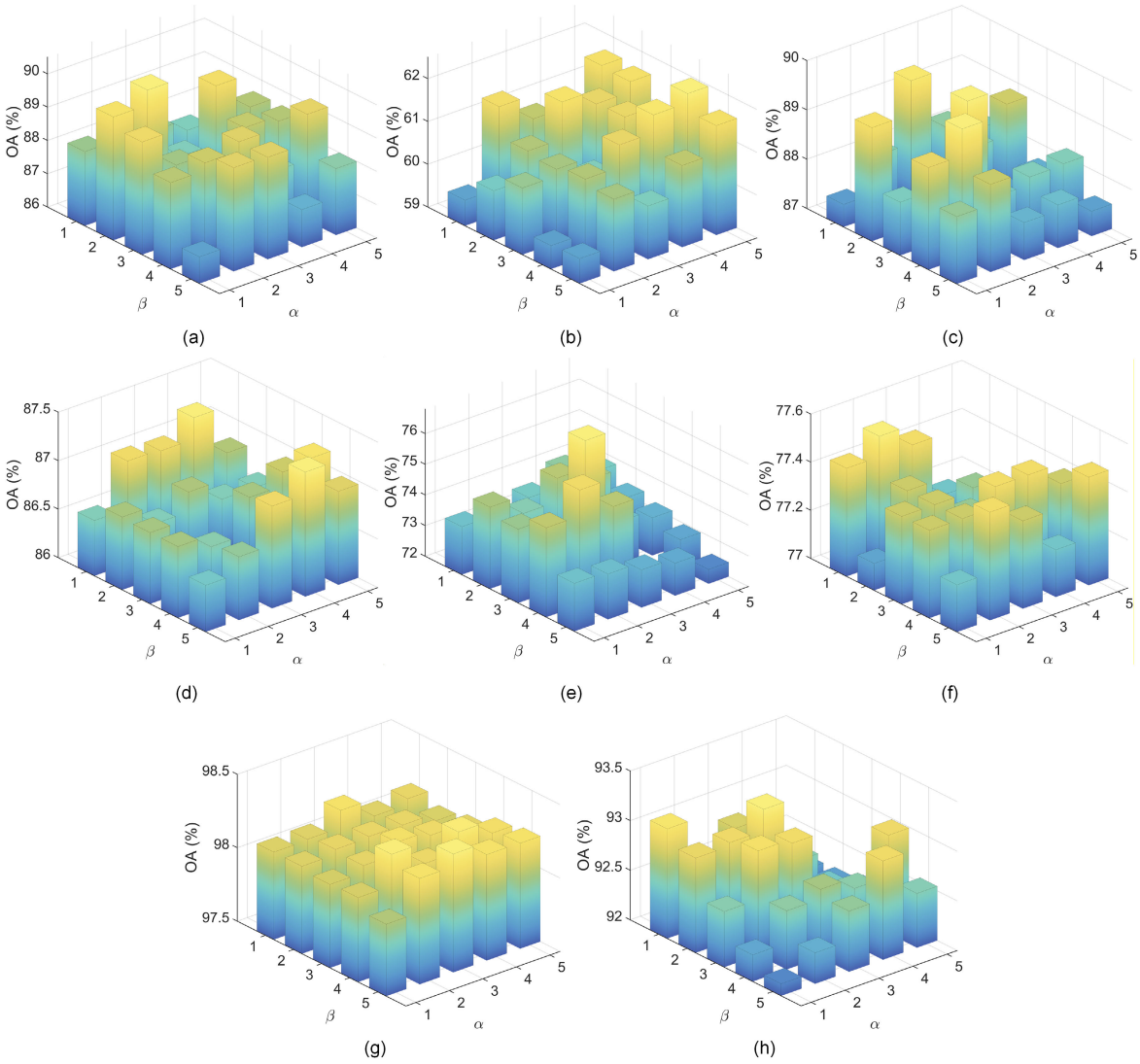


Fig. 9. Impact of different combinations of parameters α and β on OA in four datasets. (a) and (b) Cases where LiDAR and HSI are missing in the HSI–LiDAR Houston2013 dataset, respectively. (c) and (d) Cases where MS and HSI are missing in the HSI–MS Houston2013 dataset, respectively. (e) and (f) Cases where MS and HSI are missing in the HSI–MS QAI dataset, respectively. (g) and (h) Cases where LiDAR and HSI are missing in the HSI–LiDAR Trento dataset, respectively.

V. Discussion

A. Parameter Sensitivity Analysis

1) Weight Parameter (α and β): In the first stage of our proposed two-stage learning strategy, two key parameters are introduced, α and β. These two parameters provide different weights for distillation at the prediction and intermediate levels, respectively, thus influencing the student's learning of the teacher's knowledge to some extent. Therefore, we conducted experiments on four datasets to search the optimal parameter combinations for each missing scenarios. For the QAI dataset, α and β vary from 0.1 to 0.5 in steps of 0.1, while for the other three datasets, α and β vary from 1 to 5 in steps of 1. The relevant results are shown in Fig. 9.

For the HSI–LiDAR Houston2013 dataset, when HSI is missing, $\alpha = 5$ and $\beta = 4$ is the optimal combination, and when LiDAR is missing, $\alpha = 2$ and $\beta = 2$ is the optimal combination. For the HSI–MS Houston2013 dataset, when HSI is missing, the best performance is obtained with $\alpha = 4$ and $\beta = 5$, whereas when MS is missing, the optimal setting is $\alpha = 3$ and $\beta = 3$. For the HSI–MS QAI dataset, When HSI is missing, the optimal combination is $\alpha = 0.2$ and $\beta = 0.1$, while when MS is missing, the optimal combination is $\alpha = 0.3$ and $\beta = 0.3$. For HSI–LiDAR Trento dataset, when HSI is missing, $\alpha = 3$ and $\beta = 2$ is the optimal combination, and when LiDAR is missing, $\alpha = 3$ and $\beta = 5$ is the optimal combination.

Further analysis reveals several observations. For the HSI–LiDAR combination, when HSI is missing, α generally exceeds β. This can be attributed to the pronounced modality heterogeneity between HSI and LiDAR, which leads to substantial discrepancies in their feature representations. Under this condition, the absence of the more informative HSI modality makes intermediate feature alignment more challenging, thereby increasing the reliance on prediction-level knowledge transfer and necessitating a larger weight for prediction distillation. In contrast, for the HSI–MS combination, when MS

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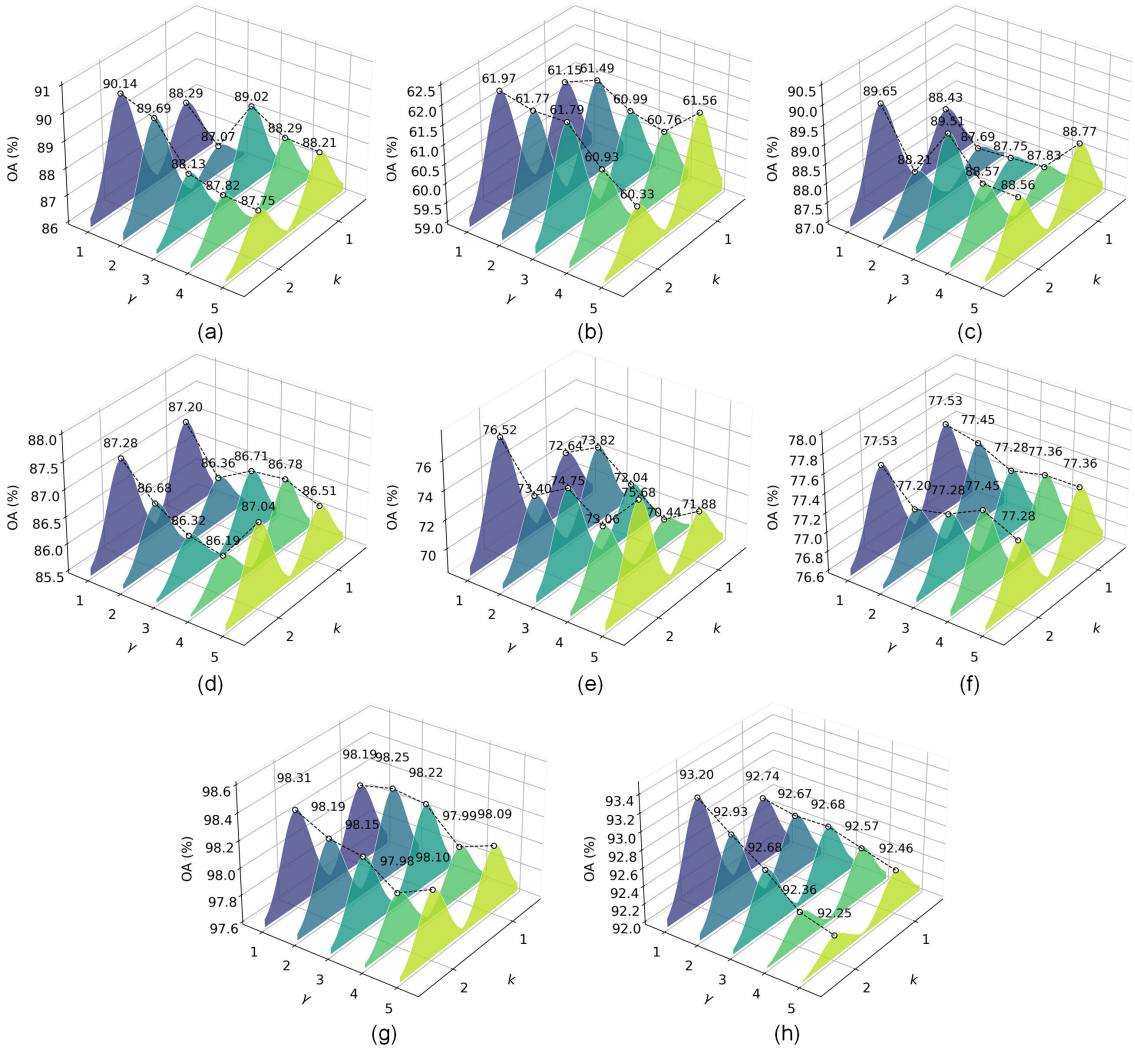


Fig. 10. Impact of different combinations of parameters k and γ on OA in four datasets. (a) and (b) Cases where LiDAR and HSI are missing in the HSI–LiDAR Houston2013 dataset, respectively. (c) and (d) Cases where MS and HSI are missing in the HSI–MS Houston2013 dataset, respectively. (e) and (f) Cases where MS and HSI are missing in the HSI–MS QAI dataset, respectively. (g) and (h) Cases where LiDAR and HSI are missing in the HSI–LiDAR Trento dataset, respectively.

is missing, α and β tend to take similar values. This may be the modality heterogeneity between HSI and MS is relatively lower, as a result, both intermediate feature distillation and prediction distillation can effectively contribute to knowledge transfer. In addition, differences among datasets in terms of spatial resolution, class diversity, and scene characteristics can further affect the selection of optimal parameter settings.

2) Grid Size k and Mean-Variance Ratio γ: When aligning the intermediate feature distribution of teacher and student, two important parameters are involved: the grid size k of spatial pyramid pooling for the feature map and the ratio γ between the mean and variance in Gaussian modeling. Similarly, we conducted experiments on four datasets to determine the optimal combination of k and γ, and the experimental results are shown in Fig. 10.

The experimental results indicate that, except for case (f) corresponding to the QAI dataset with missing HSI, where $\gamma = 1$ and $k = 1$ or 2 both yield the highest OA, the optimal OA in the remaining seven subplots under different missing-modality scenarios is achieved at $\gamma = 1$ and $k = 2$. Considering these observations, we ultimately adopt $\gamma = 1$ and $k = 2$ as the parameter configuration for all scenarios. Further analysis suggests that increasing γ amplifies the weight of the mean term, which shifts the center of the Gaussian distribution and may increase the difficulty of alignment. Meanwhile, setting $k = 2$ enables a finer partitioning of the feature maps, enhancing their representational capacity and facilitating more effective information transfer.

B. Ablation Analysis

In this section, we investigate the effectiveness of several components of the proposed method, including the hyperspectral encoder, the FW module, the DAD strategy, and the analysis of fine-tuning. To this end, we conducted several ablation experiments on the HSI–LiDAR Houston2013 dataset.

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TABLE IV

PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE HSI–MS HOUSTON2013 DATASET UNDER MISSING HSI OR MS MODALITY. BOLD AND UNDERLINED VALUES INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Method	Training	Testing	OA (%)	AA (%)	$\kappa \times 100$
Baseline					
H-M-Net[1]	HSI+MS	HSI+MS	89.95	91.23	89.10
H-Net[2]	HSI+MS	HSI	60.10	60.59	56.62
M-Net[3]	HSI+MS	MS	31.54	30.74	26.08
W/o[6]MS modality					
DGDNet	HSI+MS	HSI	83.57	86.11	82.27
mmFormer	HSI+MS	HSI	84.73	87.10	83.51
ShaSpec	HSI+MS	HSI	73.40	78.03	71.22
MSHNet	HSI+MS	HSI	85.13	87.08	83.92
ProtoKD	HSI+MS	HSI	83.77	85.97	82.47
PMKL	HSI+MS	HSI	85.18	87.00	83.98
CLKD	HSI+MS	HSI	84.58	86.84	83.33
Ours (DAD)[4]	HSI+MS	HSI	<u>89.65</u>	<u>90.64</u>	<u>88.76</u>
Ours (DAD+AF)[5]	HSI+MS	HSI	**91.05**	**91.56**	**90.28**
W/o[6]HSI modality					
DGDNet	HSI+MS	MS	85.37	88.27	84.15
mmFormer	HSI+MS	MS	85.90	88.72	84.77
ShaSpec	HSI+MS	MS	74.22	76.79	72.27
MSHNet	HSI+MS	MS	86.78	89.42	85.65
ProtoKD	HSI+MS	MS	86.85	89.48	85.83
PMKL	HSI+MS	MS	86.37	89.12	85.25
CLKD	HSI+MS	MS	86.26	89.03	85.08
Ours (DAD)[4]	HSI+MS	MS	<u>87.28</u>	<u>89.82</u>	<u>86.24</u>
Ours (DAD+AF)[5]	HSI+MS	MS	**87.40**	**89.91**	**86.36**

[1] "H-M-Net" refers to conventional multimodal model that uses HSI and MS for both training and inference.
[2] "H-Net" refers to conventional multimodal model that uses HSI and MS for training, HSI for inference.
[3] "M-Net" refers to conventional multimodal model that uses HSI and MS for training, MS for inference.
[4] "DAD" denotes Distribution Alignment Distillation.
[5] "AF" denotes Adapter Fine-Tuning.
[6] "W/o" indicates the modality missing during testing.

TABLE V

PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE HSI–MS QAI-MANGROVE DATASET UNDER MISSING HSI OR MS MODALITY. BOLD AND UNDERLINED VALUES INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Method	Training	Testing	OA (%)	AA (%)	$\kappa \times 100$
Baseline					
H-M-Net[1]	HSI+MS	HSI+MS	77.62	65.07	68.07
H-Net[2]	HSI+MS	HSI	57.94	57.02	45.58
M-Net[3]	HSI+MS	MS	63.51	28.20	37.12
W/o[6]MS modality					
DGDNet	HSI+MS	HSI	67.23	47.15	53.98
mmFormer	HSI+MS	HSI	71.03	46.38	58.67
ShaSpec	HSI+MS	HSI	67.23	42.42	53.26
MSHNet	HSI+MS	HSI	68.24	50.29	55.09
ProtoKD	HSI+MS	HSI	64.61	45.10	51.21
PMKL	HSI+MS	HSI	65.12	41.54	50.32
CLKD	HSI+MS	HSI	63.85	45.96	49.37
Ours (DAD)[4]	HSI+MS	HSI	**76.52**	<u>58.81</u>	**65.89**
Ours (DAD+AF)[5]	HSI+MS	HSI	<u>75.76</u>	**59.15**	<u>65.46</u>
W/o[6]HSI modality					
DGDNet	HSI+MS	MS	<u>78.38</u>	<u>63.00</u>	<u>68.45</u>
mmFormer	HSI+MS	MS	76.44	57.97	66.88
ShaSpec	HSI+MS	MS	77.62	55.55	67.25
MSHNet	HSI+MS	MS	72.55	59.05	61.61
ProtoKD	HSI+MS	MS	75.34	62.66	65.80
PMKL	HSI+MS	MS	73.65	54.85	63.37
CLKD	HSI+MS	MS	71.37	54.39	60.57
Ours (DAD)[4]	HSI+MS	MS	77.53	62.05	68.23
Ours (DAD+AF)[5]	HSI+MS	MS	**78.55**	**65.45**	**69.72**

[1] "H-M-Net" refers to conventional multimodal model that uses HSI and MS for both training and inference.
[2] "H-Net" refers to conventional multimodal model that uses HSI and MS for training, HSI for inference.
[3] "M-Net" refers to conventional multimodal model that uses HSI and MS for training, MS for inference.
[4] "DAD" denotes Distribution Alignment Distillation.
[5] "AF" denotes Adapter Fine-Tuning.
[6] "W/o" indicates the modality missing during testing.

1) 3-D Convolution in Hyperspectral Encoder: We replaced the 3-D convolution with 2-D convolution used for other non-HSI modalities. Table VI shows that the model performance decreased after the replacement, regardless of whether HSI or LiDAR is missing. Specifically, when HSI is missing, the OA decreased by 0.29% compared with the baseline; when LiDAR is missing, the OA decreased more significantly, by 3.38%. These decreases indicate that the 3-D convolution used in this study can effectively handle HSI band redundancy and extract rich spectral information, enhancing the model's utilization of HSI modality and improving discriminative ability.

2) Versatility of Hyperspectral Encoder: The results in Table VI demonstrate that the hyperspectral encoder contributes to the performance improvement of the proposed method. To further evaluate its effectiveness across different methods, the HSI encoder in the aforementioned competing methods is replaced with the hyperspectral encoder under the same experimental settings. The corresponding results are reported in Table VII. Compared with the counterparts without the hyperspectral encoder, taking OA as an example, the performance gains of different methods range from 0.24% to 2.75% when the LiDAR modality is missing, while the OA improvement ranges from 0.05% to 0.98% when the HSI modality is missing. These results indicate that the hyperspectral encoder can effectively enhance the exploitation of HSI features in different methods, leading to consistent performance gains and further validating its generality.

TABLE VI

PERFORMANCE COMPARISON OF DIFFERENT COMPONENTS ON THE HSI–LIDAR HOUSTON2013 DATASET UNDER MISSING HSI OR LIDAR MODALITY. BOLD VALUES INDICATE THE BEST RESULTS

Method	Training	Testing	OA (%)	AA (%)	$\kappa \times 100$
Baseline[1]	HSI+LiDAR	LiDAR	**61.97**	**64.14**	**58.90**
W/o FW[2]	HSI+LiDAR	LiDAR	61.74	63.63	58.64
W/o 3D Conv[3]	HSI+LiDAR	LiDAR	61.68	63.27	58.60
W/o FW+3D Conv[4]	HSI+LiDAR	LiDAR	61.29	63.27	58.18
Baseline[1]	HSI+LiDAR	HSI	**90.15**	**91.04**	**89.30**
W/o FW[2]	HSI+LiDAR	HSI	89.60	90.62	88.71
W/o 3D Conv[3]	HSI+LiDAR	HSI	86.77	88.85	85.66
W/o FW+3D Conv[4]	HSI+LiDAR	HSI	86.73	88.19	85.63

[1] "Baseline" refers to the first stage result in our proposed method.
[2] "W/o FW" denotes the removal of the Feature Weighting.
[3] "W/o 3D Conv" denotes replacing 3D convolution with 2D convolution.
[4] "W/o FW+3D Conv" denotes removing the Feature Weighting and replacing 3D convolution with 2D convolution.

3) Feature Weighting: To evaluate the effectiveness of FW, we removed this component to observe changes in the model's performance. As shown in Table VI, removing FW resulted in

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TABLE VII

PERFORMANCE COMPARISON OF DIFFERENT METHODS WITH THE HYPERSPECTRAL ENCODER ON THE HSI–LIDAR HOUSTON2013 DATASET UNDER MISSING HSI OR LIDAR MODALITY

Method	Training	Testing	OA (%)	AA (%)	$\kappa \times 100$
W/o[1] LiDAR modality					
DGDNet	HSI+LiDAR	HSI	84.71	86.59	83.44
mmFormer	HSI+LiDAR	HSI	87.71	89.12	86.67
ShaSpec	HSI+LiDAR	HSI	82.27	83.85	80.80
MSHNet	HSI+LiDAR	HSI	87.76	89.11	86.71
ProtoKD	HSI+LiDAR	HSI	87.39	88.85	86.31
PMKL	HSI+LiDAR	HSI	86.08	87.72	84.89
CLKD	HSI+LiDAR	HSI	86.63	87.62	85.48
W/o[1] HSI modality					
DGDNet	HSI+LiDAR	LiDAR	60.24	62.02	57.05
mmFormer	HSI+LiDAR	LiDAR	56.38	58.47	52.97
ShaSpec	HSI+LiDAR	LiDAR	49.37	48.22	45.07
MSHNet	HSI+LiDAR	LiDAR	61.04	62.97	57.91
ProtoKD	HSI+LiDAR	LiDAR	60.86	63.09	57.73
PMKL	HSI+LiDAR	LiDAR	60.42	61.65	57.27
CLKD	HSI+LiDAR	LiDAR	55.87	56.52	52.36

[1] "W/o" indicates the modality missing during testing.

a 0.23% decrease in OA when HSI is missing, and a 0.55% decrease when LiDAR is missing. Furthermore, removing FW after replacing the 3-D convolution further degraded the model's performance. These results validate the effectiveness of FW, demonstrating its positive role in enabling student to stably learn complete multimodal information from the teacher under missing modalities.

4) Effect of DAD Strategy: To study the effect of our proposed distillation strategy, we applied different distillation methods to our first-stage model to compare the performance between the methods. Specifically, we first ablate our method by removing the logit and feature distillation components, respectively. Then, we use distillation methods involved in the comparison methods, such as ProtoKD, PMKL, JAD in MSHNet, and DGD in DGDNet. Note that CLKD uses the same logit distillation method as our method after removing the feature component. In addition, the widely used feature value matching method mse, and three representative distribution alignment methods, namely, Sinkhorn distance (SD), central moment discrepancy (CMD), and correlation alignment (CORAL), are adopted for comparison.

As shown in Table VIII, the model performance decreased to varying degrees after removing either the logit or feature component. Notably, when HSI is missing, the classificationaccuracy of "W/o Logit" decreased more than that of "W/o Feature," while when LiDAR is missing, the difference in classification accuracy is only slight. This is consistent with the conclusion in the previous parameter sensitivity analysis that the model relies more on logit distillation when HSI is missing. Compared with the distillation strategies employed by the competing methods, the proposed method consistently achieves superior classification performance. Furthermore, the proposed method consistently outperforms both direct feature value matching and the three representative distribution alignment methods. These results demonstrate the effectiveness and superiority of transferring knowledge by minimizing the Wasserstein distance between Gaussian distributions.

TABLE VIII

PERFORMANCE COMPARISON OF DIFFERENT DISTILLATION STRATEGIES ON THE HSI–LIDAR HOUSTON2013 DATASET UNDER MISSING HSI OR LIDAR MODALITY. BOLD VALUES INDICATE THE BEST RESULTS

Method	Training	Testing	OA (%)	AA (%)	$\kappa \times 100$
Baseline[1]	HSI+LiDAR	LiDAR	**61.97**	**64.14**	**58.90**
W/o Logit[2]	HSI+LiDAR	LiDAR	54.78	57.49	51.40
W/o Feature[3]	HSI+LiDAR	LiDAR	60.85	62.64	57.72
MSE	HSI+LiDAR	LiDAR	53.27	55.84	49.68
Logit+MSE	HSI+LiDAR	LiDAR	61.15	63.20	58.03
Logit+SD[4]	HSI+LiDAR	LiDAR	61.13	63.22	58.03
Logit+CMD[5]	HSI+LiDAR	LiDAR	60.81	62.76	57.66
Logit+CORAL[6]	HSI+LiDAR	LiDAR	59.50	60.93	56.30
ProtoKD	HSI+LiDAR	LiDAR	61.47	63.60	58.42
PMKL	HSI+LiDAR	LiDAR	61.83	63.00	58.75
JAD	HSI+LiDAR	LiDAR	61.85	63.28	58.76
DGD	HSI+LiDAR	LiDAR	60.70	62.13	57.53
Baseline[1]	HSI+LiDAR	HSI	**90.15**	**91.04**	**89.30**
W/o Logit[2]	HSI+LiDAR	HSI	88.75	89.89	87.80
W/o Feature[3]	HSI+LiDAR	HSI	88.34	89.68	87.35
MSE	HSI+LiDAR	HSI	87.92	89.22	86.88
Logit+MSE	HSI+LiDAR	HSI	88.68	89.96	87.71
Logit+SD[4]	HSI+LiDAR	HSI	88.92	90.09	87.97
Logit+CMD[5]	HSI+LiDAR	HSI	84.91	86.99	83.66
Logit+CORAL[6]	HSI+LiDAR	HSI	88.49	89.75	87.52
ProtoKD	HSI+LiDAR	HSI	88.69	89.95	87.74
PMKL	HSI+LiDAR	HSI	87.32	88.71	86.24
JAD	HSI+LiDAR	HSI	89.66	90.70	88.79
DGD	HSI+LiDAR	HSI	89.74	90.67	88.87

[1] "Baseline" refers to the first stage result in our proposed method.
[2] "W/o Logit" refers to the removal of logit distillation.
[3] "W/o Feature" refers to the removal of feature distillation.
[4] "SD" refers to the Sinkhorn Distance.
[5] "CMD" refers to the Central Moment Discrepancy.
[6] "CORAL" refers to the CORrelation ALignment.

5) Analysis of Fine-Tuning Strategies and Positions: Fine-tuning serves as a core component of the second-stage learning, and the choice of fine-tuning strategy and position directly affects the final performance. To determine the optimal fine-tuning scheme, three representative fine-tuning methods, namely, AFT, affine transformation (AT), and low-rank adaptation (LoRA), are selected and investigated at two different positions. Specifically, the two positions are located at the interfaces between the encoder and FW, and FW and the Classifier, respectively. As indicated by the ablation results in Table IX, simultaneously introducing AFT at both positions yields the best performance when either HSI modality or LiDAR modality is missing. Therefore, this configuration is adopted as the final fine-tuning strategy.

C. Feature Separability

To qualitatively evaluate the effectiveness of our method under missing-modality scenarios, we employ t-SNE to visualize the separability of the final-layer features before classification on the HSI–LiDAR Houston2013 dataset. Specifically, 140 samples are randomly selected for each class, yielding a total of 2100 samples, and t-SNE is used to project the high-dimensional features into a 2-D space for visualization. Figs. 11 and 12 present the visualization results for scenarios where LiDAR and HSI modalities are missing, respectively. When LiDAR is absent, our method demonstrates

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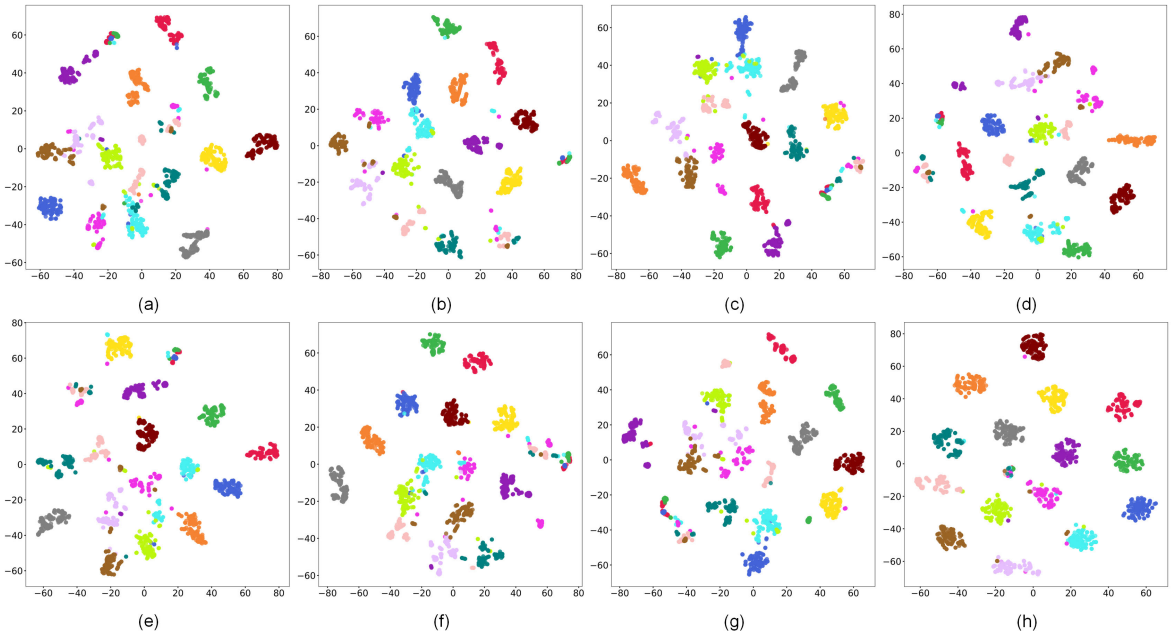


Fig. 11. t-SNE results of different methods on the HSI–LiDAR Houston2013 dataset when LiDAR is missing. (a) DGDNet. (b) mmFormer. (c) ShaSpec. (d) MSHNet. (e) ProtoKD. (f) PMKL. (g) CLKD. (h) Ours.

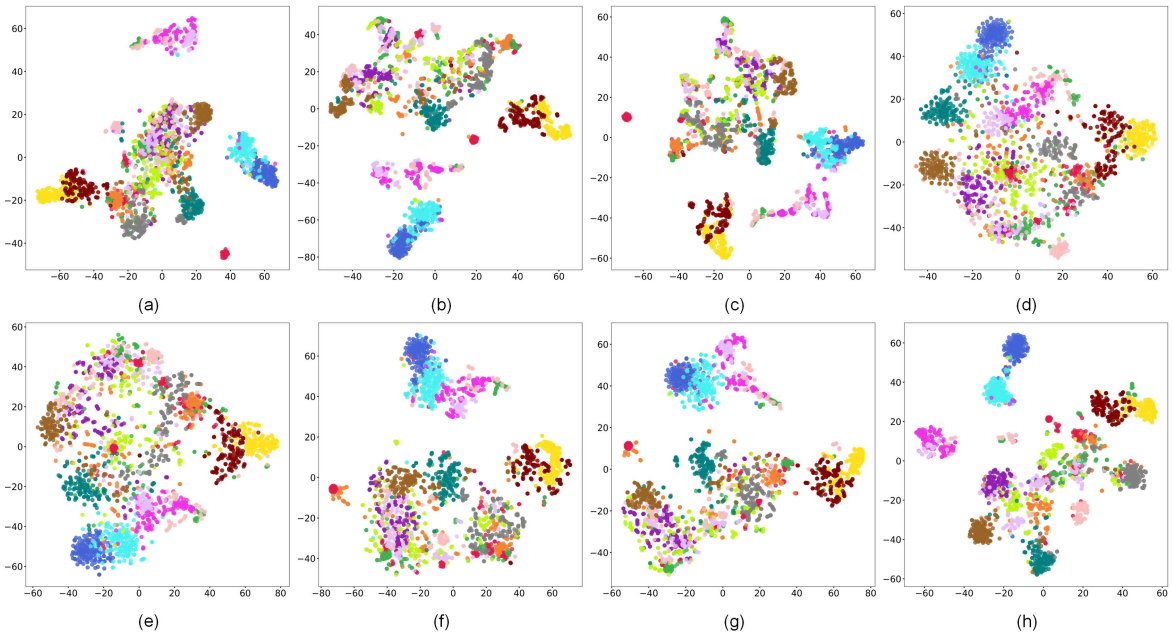


Fig. 12. t-SNE results of different methods on the HSI–LiDAR Houston2013 dataset when HSI is missing. (a) DGDNet. (b) mmFormer. (c) ShaSpec. (d) MSHNet. (e) ProtoKD. (f) PMKL. (g) CLKD. (h) Ours.

the highest class separability, with most classes forming distinct clusters and exhibiting clear boundaries. In contrast, when HSI is missing, the limited discriminative capability of LiDAR leads to less pronounced cluster separation across all methods. Nevertheless, our method still achieves the best class separability, whereas the other methods show relatively low interclass separability, with clusters from different categories substantially overlapping. These visualizations provide qualitative evidence that our approach effectively preserves discriminative information and mitigates the impact of missing modalities.

D. CKA Similarity Between Missing and Complete Modalities

To quantify the discrepancy between representations learned by different methods under missing and complete modalities, we use centered kernel alignment (CKA) [60] to calculate the similarity between the last layer feature before classification in

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TABLE IX

PERFORMANCE COMPARISON OF DIFFERENT FINE-TUNING STRATEGIES AND POSITIONS ON THE HSI–LIDAR HOUSTON2013 DATASET UNDER MISSING HSI OR LIDAR MODALITY. BOLD VALUES INDICATE THE BEST RESULTS

Missing	Method	E-F[1]	F-C[2]	OA (%)	AA (%)	$\kappa \times 100$
HSI	AFT[3]	✓	✓	**62.15**	**64.38**	**59.12**
		✓		61.38	63.29	58.26
			✓	61.50	63.43	58.40
	AT[4]	✓	✓	62.01	63.51	58.95
		✓		62.06	63.14	58.98
			✓	62.02	63.42	58.94
	LoRA[5]	✓	✓	62.03	63.71	59.00
		✓		61.84	63.66	58.78
			✓	62.04	63.57	59.00
LiDAR	AFT[3]	✓	✓	**90.63**	**91.42**	**89.82**
		✓		88.45	89.64	87.45
			✓	87.62	88.91	86.56
	AT[4]	✓	✓	88.21	89.34	87.19
		✓		89.49	90.39	88.59
			✓	88.64	89.85	87.67
	LoRA[5]	✓	✓	90.60	91.29	89.80
		✓		89.69	90.70	88.82
			✓	90.31	91.14	89.48

[1] "E-F" denotes the position between the Encoder and Feature Weighting modules.
[2] "F-C" denotes the position between the Feature Weighting and Classifier.
[3] "AFT" denotes Adapter Fine-Tuning.
[4] "AT" denotes Affine Transformation.
[5] "LoRA" denotes Low-Rank Adaptation.

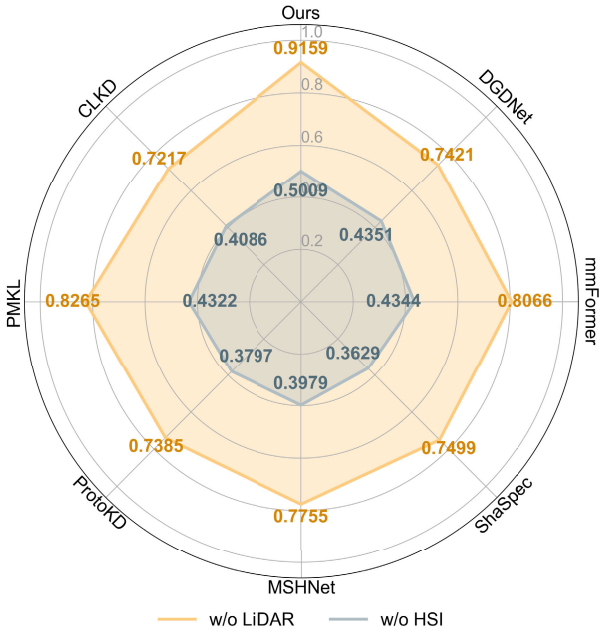


Fig. 13. CKA similarity of different methods on the HSI–LiDAR Houston2013 dataset when LiDAR or HSI is missing.

the missing and complete modalities. This metric can quantify the structural similarity between two high-dimensional feature representations. Consequently, it provides a robust assessment of how well the features extracted under missing modalities align with those from complete modalities. Fig. 13 shows the CKA similarity calculated by each method under two missing-modality scenarios. Our method achieves 0.5009 for missing HSI and 0.9159 for missing LiDAR. In both cases, our method achieves the highest similarity, indicating that our method enables the model to learn representations under missing modalities with minimal discrepancy from the complete modalities.

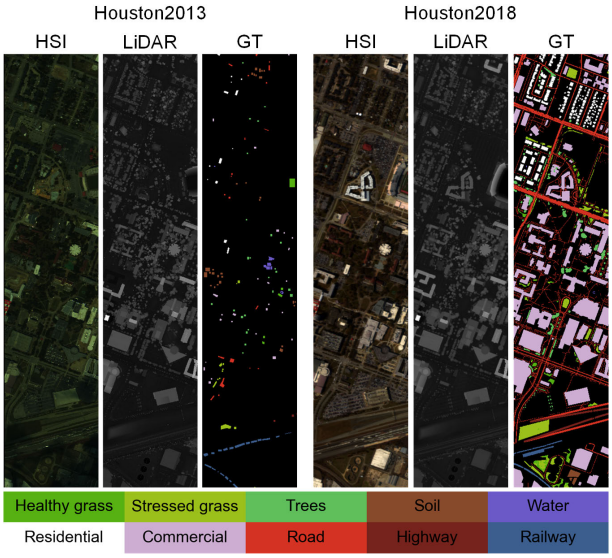


Fig. 14. Multimodal images and GT maps of Houston datasets.

TABLE X

NUMBER OF SAMPLES FOR CROSS-TEMPORAL DATASETS

Class		Number of Samples	
No.	Name	Houston2013 (Source)	Houston2018 (Target)
1	Healthy grass	345	1353
2	Stressed grass	365	4888
3	Trees	365	2766
4	Soil	650	448
5	Water	285	22
6	Residential	319	5347
7	Commercial	408	32459
8	Road	443	17746
9	Highway	46	1574
10	Railway	729	956
Total		3955	67559

E. Missing Modalities With Temporal Variation

Unlike the previous same-period experiments, the Houston datasets acquired in 2013 and 2018 were used for cross-temporal training and testing. This more challenging and realistic setting was adopted to further evaluate the robustness of CSKLNet under missing modalities.

The Houston 2013 dataset contains a 144-band HSI covering 380–1050 nm at a spatial resolution of 2.5 m, whereas the Houston 2018 dataset contains a 48-band HSI over the same wavelength range at a spatial resolution of 1 m. We extracted 48 spectral bands (wavelength range 380–1050 nm) from the Houston 2013 scene corresponding to the Houston 2018 scene, and selected the overlapping area of 209×955. There are ten consistent classes in their scene; the classes and the number of samples are listed in Table X. In addition,

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TABLE XI

PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE CROSS-TEMPORAL HOUSTON DATASETS UNDER MISSING HSI OR LIDAR MODALITY. BOLD AND UNDERLINED VALUES INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Method	Training	Testing	OA (%)	AA (%)	$\kappa \times 100$
W/o[3]LiDAR modality					
DGDNet	HSI+LiDAR	HSI	59.12	52.04	46.57
mmFormer	HSI+LiDAR	HSI	44.44	42.59	32.66
ShaSpec	HSI+LiDAR	HSI	47.94	52.18	33.46
MSHNet	HSI+LiDAR	HSI	56.84	71.24	47.11
ProtoKD	HSI+LiDAR	HSI	59.07	62.37	46.79
PMKL	HSI+LiDAR	HSI	52.78	69.57	43.09
CLKD	HSI+LiDAR	HSI	52.56	69.90	43.20
Ours (DAD)[1]	HSI+LiDAR	HSI	<u>71.45</u>	<u>71.89</u>	<u>62.07</u>
Ours (DAD+AF)[2]	HSI+LiDAR	HSI	**71.90**	**73.76**	**62.78**
W/o[3]HSI modality					
DGDNet	HSI+LiDAR	LiDAR	47.26	32.33	34.04
mmFormer	HSI+LiDAR	LiDAR	48.80	22.09	28.78
ShaSpec	HSI+LiDAR	LiDAR	39.60	25.64	17.43
MSHNet	HSI+LiDAR	LiDAR	57.67	**44.06**	44.40
ProtoKD	HSI+LiDAR	LiDAR	54.10	30.91	39.47
PMKL	HSI+LiDAR	LiDAR	51.97	32.94	37.86
CLKD	HSI+LiDAR	LiDAR	55.26	<u>41.32</u>	41.59
Ours (DAD)[1]	HSI+LiDAR	LiDAR	<u>58.55</u>	35.13	<u>44.98</u>
Ours (DAD+AF)[2]	HSI+LiDAR	LiDAR	**59.13**	35.53	**45.77**

[1] "DAD" denotes Distribution Alignment Distillation.
[2] "AF" denotes Adapter Fine-Tuning.
[3] "W/o" indicates the modality missing during testing.

their visualization maps are shown in Fig. 14. In the cross-temporal setting, the Houston 2013 dataset with complete multimodal data is regarded as the source domain, whereas the Houston 2018 dataset with missing modalities during inference is regarded as the target domain. In addition, for stable training, the learning rate of all methods was set to $1e^{-4}$, while all other parameters remained the same as in the previous settings.

Table XI presents the classification performance of CSKLNet and the competing methods in the cross-temporal missing modalities experiments. When the LiDAR modality is missing in the target domain, CSKLNet achieves improvements across all evaluation metrics. Taking OA as an example, the improvement ranges from 12.78% to 27.46% compared with the competing methods. When the HSI modality is missing in the target domain, CSKLNet outperforms the competing methods on all metrics except AA, with OA improvements ranging from 1.46% to 19.53%. Further analysis reveals that CSKLNet achieves larger performance gains when only the HSI modality is available in the target domain. This may be attributed to the larger distribution discrepancy between HSI data acquired in different years, whereas LiDAR data are relatively less affected by temporal variations. Benefiting from the advantage of the Wasserstein distance in mitigating distribution discrepancies, CSKLNet can obtain greater performance improvements in this scenario. However, compared with the same-temporal setting, all methods exhibited performance degradation to varying degrees, highlighting a challenge that merits further investigation in future work. Overall, CSKLNet demonstrates stronger robustness and generalizability under cross-temporal missing modalities conditions, highlighting its greater potential for practical applications.

VI. CONCLUSION

In this work, we propose a novel network, called CSKLNet, to address multimodal remote sensing image classification with missing modalities. Through a two-stage learning framework, CSKLNet effectively integrates modality-common knowledge and modality-specific knowledge, providing sufficient informative cues for classification when modalities are missing. Specifically, CSKLNet first employs DAD to transfer comprehensive multimodal information from the teacher network to the student network, thereby enabling the student to learn common knowledge across modalities under missing modalities. Subsequently, AFT is applied to the distilled student network to further capture and integrate modality-specific information from the available modalities. Moreover, beyond conventional distillation approaches, we further align feature distributions by minimizing the Wasserstein distance and incorporate classification boundary constraints during the distillation process. This effectively alleviates representation heterogeneity and facilitates the transfer of modality-common knowledge from the multimodal teacher. Finally, extensive experimental results on four datasets demonstrate the effectiveness of the proposed method.

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